

A quantitative replica-symmetric bound in strict AT region

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Abstract

We consider the Sherrington–Kirkpatrick model with inverse temperature $\beta > 0$ and deterministic external field $h > 0$. Let q be the replica-symmetric fixed point:

$$q = \mathbb{E} \tanh^2(h + \beta\sqrt{q}Z), \quad Z \sim N(0, 1).$$

We prove that, uniformly on compact subsets of the strict de Almeida–Thouless region:

$$\beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta\sqrt{q}Z) < 1,$$

the overlap satisfies the concentration:

$$\mathbb{E} \langle (R_{12} - q)^2 \rangle = O(N^{-1}).$$

As a consequence, we obtain an $O(N^{-1})$ replica-symmetric free-energy correction and identify the finite-volume replicon susceptibility. Unlike the approach of Lopatto [17], our proof does not use the identification of the limiting free energy with the Parisi variational formula.

1 Model and main results

The Sherrington–Kirkpatrick model is the basic mean-field model of a disordered Ising system. In a nonzero external field, its replica-symmetric order parameter is determined by a scalar fixed-point equation, while the de Almeida–Thouless condition describes the stability of that solution against two-replica perturbations. The strict inequality in the AT condition is therefore expected to imply that the overlap has fluctuations of order $N^{-1/2}$ and that the replica-symmetric free energy is accurate up to order N^{-1} . Our purpose is to prove these claims uniformly on compact subsets of the strict AT region.

1.1 Model and the strict AT region

Let $N \in \mathbb{N}$ and $\Sigma_N := \{-1, 1\}^N$. For $\beta > 0$ and $h > 0$, let

$$H_N(\sigma) := \frac{\beta}{\sqrt{2N}} \sum_{i,j=1}^N g_{ij} \sigma_i \sigma_j + h \sum_{i=1}^N \sigma_i, \quad \sigma \in \Sigma_N$$

where $(g_{ij})_{1 \leq i,j \leq N}$ are independent standard normal random variables.

Let $q(\beta, h)$ be the unique solution of

$$q = \mathbb{E} \tanh^2(h + \beta\sqrt{q}Z), \quad Z \sim N(0, 1), \tag{1.1}$$

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and write $q := q(\beta, h)$. Existence and uniqueness for $h > 0$ are standard; see, for example, [19, Proposition 1.3.8]. Continuity follows directly from uniqueness: if $(\beta_n, h_n) \rightarrow (\beta, h)$, every convergent subsequence of $q(\beta_n, h_n) \in [0, 1]$ has, by dominated convergence in (1.1), a limit solving the fixed-point equation at (β, h) . Uniqueness identifies that limit with $q(\beta, h)$, so the entire sequence converges. Set

$$r := \mathbb{E} \tanh^4(h + \beta\sqrt{q} Z), \quad (1.2)$$

and let

$$\alpha := \beta^2(1 - 2q + r) = \beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta\sqrt{q} Z). \quad (1.3)$$

We work on a fixed compact set

$$\mathcal{K} \Subset \{(\beta, h) : \beta > 0, h > 0, \alpha(\beta, h) < 1\}. \quad (1.4)$$

Thus, for some $\delta_{\mathcal{K}} > 0$,

$$1 - \alpha \geq \delta_{\mathcal{K}} \quad \text{on } \mathcal{K}. \quad (1.5)$$

The maps q, r, α are therefore continuous on $\mathcal{K}$. In particular, compactness and $h > 0$ imply

$$0 < q_{\mathcal{K}} := \inf_{(\beta, h) \in \mathcal{K}} q \leq \sup_{(\beta, h) \in \mathcal{K}} q < 1. \quad (1.6)$$

Indeed, $q = 0$ would contradict (1.1) and $h > 0$; $q < 1$ holds because $\tanh^2(h + \beta\sqrt{q} Z) < 1$.

1.2 The replica-symmetric smart path

For $0 \leq s \leq 1$, define the smart-path Hamiltonian

$$H_{N,s}(\sigma) := \frac{\beta\sqrt{s}}{\sqrt{2N}} \sum_{i,j=1}^N g_{ij} \sigma_i \sigma_j + \sum_{i=1}^N (h + \beta\sqrt{(1-s)q} z_i) \sigma_i, \quad (1.7)$$

where (z_i) is an independent standard Gaussian family. We write $\langle \cdot \rangle_s$ for the Gibbs expectation, i.e.

$$\langle F \rangle_s := \frac{1}{Z_{N,s}} \sum_{\sigma \in \Sigma_N} F(\sigma) e^{H_{N,s}(\sigma)}$$

for a function F on Σ_N , where $Z_{N,s} := \sum_{\sigma \in \Sigma_N} e^{H_{N,s}(\sigma)}$. For replicas $\sigma^1, \sigma^2, \dots$ sampled independently from the Gibbs measure for fixed disorder, put

$$R_{ab} := \frac{1}{N} \sum_{i=1}^N \sigma_i^a \sigma_i^b.$$

Set

$$\nu_s[f] := \mathbb{E} \langle f \rangle_s, \quad Q_{ab} := R_{ab} - q.$$

We also write

$$\phi_N(s) := \frac{1}{N} \mathbb{E} \log Z_{N,s}. \quad (1.8)$$

1.3 Main result

Define

$$A_s := \nu_s[Q_{12}^2], \quad B_s := \nu_s[Q_{12}Q_{13}], \quad C_s := \nu_s[Q_{12}Q_{34}]. \quad (1.9)$$

Theorem 1.1. *There is a constant $C_{\mathcal{K}} < \infty$ such that, for every $N \geq 1$,*

$$\sup_{\substack{(\beta, h) \in \mathcal{K} \\ 0 \leq s \leq 1}} N \nu_s[Q_{12}^2] \leq C_{\mathcal{K}}.$$

Consequently, if

$$\phi_N(\beta, h) := \frac{1}{N} \mathbb{E} \log \sum_{\sigma \in \Sigma_N} e^{H_N(\sigma)}$$

and

$$\phi^{\text{RS}}(\beta, h) := \log 2 + \mathbb{E} \log \cosh(h + \beta \sqrt{q} Z) + \frac{\beta^2}{4} (1 - q)^2,$$

then uniformly on $\mathcal{K}$,

$$0 \leq \phi^{\text{RS}}(\beta, h) - \phi_N(\beta, h) \leq \frac{C_{\mathcal{K}}}{N}.$$

Moreover, uniformly for $(\beta, h, s) \in \mathcal{K} \times [0, 1]$,

$$N(A_s - 2B_s + C_s) = \frac{\alpha}{\beta^2(1 - s\alpha)} + o_{\mathcal{K}}(1).$$

Here and below, $o_{\mathcal{K}}(1) \rightarrow 0$ as $N \rightarrow \infty$, uniformly over the compact set $\mathcal{K}$.

1.4 A brief history and previous research

Spin glasses were introduced to describe magnetic materials in which the interactions between microscopic spins are both disordered and frustrated. Experiments on dilute magnetic alloys in the 1970s showed behavior very different from that of ordinary ferromagnets: a sharp freezing transition, very slow relaxation, and a strong dependence on the history of the sample. These findings suggested that the low-temperature phase could not be described by a single equilibrium state of the usual kind, and they led to probabilistic models with quenched disorder.

One of the first such models was proposed by Edwards and Anderson [4]. In their model, spins sit on the vertices of a finite-dimensional lattice and interact through random nearest-neighbor couplings. This formulation is physically natural, but its mathematical analysis is very hard. The local interactions compete with each other because of frustration, and the quenched disorder removes the spatial regularity that classical arguments rely on. As a result, even basic questions about the low-temperature phase remain difficult in finite dimensions.

A decisive simplification was proposed by Sherrington and Kirkpatrick [11]. They replaced the short-range geometry of the Edwards–Anderson model by a mean-field interaction, in which every spin interacts weakly with every other spin. The resulting model, now called the Sherrington–Kirkpatrick (SK) model, keeps both disorder and frustration but is much easier to study by analytic methods. Using the replica method, Sherrington and Kirkpatrick derived a formula for the limiting free energy under the assumption of replica symmetry. The replica method is not rigorous: it starts from identities for integer moments of the partition function and then uses an analytic continuation in the number of replicas. More importantly, the replica-symmetric formula gave a negative entropy at low temperature, which is unphysical. It was soon understood that the replica-symmetric ansatz could not describe the whole phase diagram.

A fundamental step was taken by de Almeida and Thouless [3], who studied the local stability of the replica-symmetric solution. Their computation predicted that the replica-symmetric solution is stable when

$$\beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta \sqrt{q} Z) \leq 1, \quad (1.10)$$

and unstable when the reverse strict inequality holds. This boundary is now called the de Almeida–Thouless line, or AT line. So, the AT condition comes from the physical stability analysis of the replica-symmetric state.

The failure of replica symmetry below the AT line led Parisi to propose a much richer picture of the low-temperature phase [9, 10]. In this picture, the symmetry among replicas is broken in a hierarchical way, and the thermodynamic state is described by an order parameter that gives the distribution of overlaps between typical configurations. This led to the Parisi variational formula for the limiting free energy. The Parisi theory fixed the low-temperature problem of the replica-symmetric solution and gave a single framework that covers both the replica-symmetric and the replica-symmetry-breaking phases.

These ideas have since found use far beyond condensed matter physics. In computer science, random constraint satisfaction problems such as random K -SAT and random graph coloring show threshold phenomena similar to replica symmetry breaking, and algorithms based on the cavity method, such as survey propagation, are still among the best known methods for these problems [7, 6]. In information theory and high-dimensional statistics, the free energy of a suitable spin glass model gives the fundamental limits in error-correcting codes, compressed sensing, and inference in graphical models [8, 12]. In machine learning, the Hopfield model of associative memory is a spin glass in which the stored patterns play the role of quenched disorder [5, 2], and spin glass methods have been used to study the loss landscapes and training dynamics of neural networks. More generally, the notions of frustration, energy landscapes, and metastability, first identified in spin glasses, are now standard tools in the study of complex systems, from protein folding to financial networks.

The rigorous mathematical study of the SK model developed along a different route. An early landmark was the work of Aizenman, Lebowitz, and Ruelle [1], who proved Gaussian fluctuations of the free energy at high temperature and zero external field, using a cluster expansion. At zero external field, replica symmetry is now known to hold for all $\beta < 1$, and the critical point $\beta = 1$ separates the replica-symmetric and symmetry-breaking regimes. With an external field, finding the full replica-symmetric region turned out to be much harder. Talagrand proved the replica-symmetric formula at sufficiently high temperature with the cavity method, and a simple interpolation argument due to Łatała gave another proof in the range $\beta \in [0, 1/2)$ and $h \in \mathbb{R}$. A recent work of the authors extended Łatała’s argument to a wider range including the region $\beta \leq 1$. These high-temperature results, together with the interpolation methods developed at the same time, form much of the mathematical basis for the quantitative study of the overlap.

In parallel, a different line of work dealt with the limiting free energy at all temperatures. Guerra introduced an interpolation argument that proved the appropriate Parisi bound, and Talagrand then proved the matching opposite bound. Together, these results established the Parisi formula for the SK model. This turned the study of the phase diagram from a problem based on the replica calculus into a rigorous variational problem. In particular, replica symmetry can be characterized through the minimizer of the Parisi functional, and this variational point of view has become one of the main tools for studying the AT transition.

One question remained: does the AT stability condition exactly characterize the replica-symmetric phase of the SK model with a nonzero external field? Several rigorous results proved replica symmetry in regions inside the predicted AT region, or replica symmetry breaking on the other side of

the line, but reaching the physical boundary itself remained difficult. Very recently, Lopatto [17] proved, for a positive uniform external field, that replica symmetry holds throughout the AT region (1.10) using the variational characterization of the Parisi measure. The de Almeida–Thouless prediction for the replica-symmetric side of the phase diagram is therefore now rigorously established.

The present paper gives a quantitative counterpart of this picture. Instead of asking only whether the thermodynamic limit is replica symmetric, we work under the strict AT condition and prove the quantitative estimate stated above. The strict inequality gives a positive stability margin away from the AT boundary, and our argument turns this margin into finite- N control. In this sense, our result complements the characterization of the replica-symmetric phase.

1.5 Outline of the proof: from coupled free energy to optimal overlap bound

We briefly describe the proof of the first claim in the main result; the second and third are obtained from this using a standard interpolation argument (see Section 4).

For $u \in \mathbb{R}$, define the constrained two-replica pressure by

$$p_{N,s}^{(2)}(u) := \frac{1}{2N} \mathbb{E} \log \sum_{\sigma^1, \sigma^2} \exp \left(H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2) + \frac{uN}{2} Q_{12}^2 \right) \quad (1.11)$$

The Guerra–Talagrand interpolation gives

$$p_{N,s}^{(2)}(u) \leq \mathfrak{P}_s(u) + o_{\mathcal{K}}(1) \quad (1.12)$$

uniformly in $(\beta, h) \in \mathcal{K}$, $s \in [0, 1]$, and $u \in [-1, 1]$. Here $\mathfrak{P}_s(u)$ denotes the corresponding one-step replica-symmetry-breaking formula. Using Lopatto’s argument [17], one obtains that as $u \rightarrow 0+$,

$$\mathfrak{P}_s(u) = \mathfrak{P}_s(0) + o(u). \quad (1.13)$$

Let

$$D_N(s) := \mathfrak{P}_s(0) - \phi_N(s).$$

By Gaussian integration by parts, we have for $u_0 > 0$ sufficiently small,

$$D'_N(s) = \frac{\beta^2}{4} \nu_s[Q_{12}^2] \leq \frac{\beta^2}{u_0} (p_{N,s}^{(2)}(u_0) - \phi_N(s)) \leq \frac{\beta^2}{u_0} D_N(s) + o_{\mathcal{K}}(1),$$

where we have used Jensen’s inequality with $p_{N,s}^{(2)}(u_0) - \phi_N(s) = \mathbb{E} \log \langle e^{u_0 N Q_{12}^2/2} \rangle_s$ for the first inequality, and (1.12) and (1.13) with $o_{\mathcal{K}}(1)$ being arbitrarily small for the last inequality. Since $D_N(0) = 0$, by Gronwall’s inequality, we have

$$D_N(s) = o_{\mathcal{K}}(1), \quad p_{N,s}^{(2)}(u_0) - \phi_N(s) = \frac{1}{2N} \mathbb{E} \log \langle e^{u_0 N Q_{12}^2/2} \rangle_s = o_{\mathcal{K}}(1).$$

Hence, using a standard Gaussian Lipschitz concentration inequality, one gets the preliminary overlap concentration estimate

$$\sup_{\substack{(\beta, h) \in \mathcal{K} \\ 0 \leq s \leq 1}} \nu_s(|Q_{12}| \geq \delta) \leq C_{\mathcal{K}, \delta} \exp(-c_{\mathcal{K}, \delta} N), \quad \delta > 0. \quad (1.14)$$

To obtain the N^{-1} bound, we use the cavity bootstrap developed in the high-temperature analysis of Talagrand [19, Chapter 1]: Estimating the influence of adding one spin, we reach

$$\nu_s[Q_{12}^2] \leq C_{\mathcal{K}} (\nu_s|Q_{12}|^3 + N^{-1}). \quad (1.15)$$

The preliminary concentration estimate (1.14) yields that for any fixed $0 < \delta$,

$$\begin{aligned}\nu_s|Q_{12}|^3 &\leq \delta\nu_s[Q_{12}^2] + 8\nu_s(|Q_{12}| \geq \delta) \\ &\leq \delta\nu_s[Q_{12}^2] + C_{\mathcal{K},\delta}e^{-c_{\mathcal{K},\delta}N}.\end{aligned}\tag{1.16}$$

Consequently, we have

$$\nu_s[Q_{12}^2] \leq \delta C_{\mathcal{K}}\nu_s[Q_{12}^2] + C_{\mathcal{K}}N^{-1} + C_{\mathcal{K}}C_{\mathcal{K},\delta}e^{-c_{\mathcal{K},\delta}N}.$$

Choosing $\delta = 1/(2C_{\mathcal{K}}) > 0$, we obtain

$$\sup_{\substack{(\beta,h) \in \mathcal{K} \\ 0 \leq s \leq 1}} \nu_s[Q_{12}^2] \leq \frac{C'_{\mathcal{K}}}{N},\tag{1.17}$$

with some $C'_{\mathcal{K}}$. This completes the proof of the quantitative bound for overlaps.

2 Replica-symmetric coercivity

2.1 Gaussian calculus

We record Gaussian integration by parts, which is used below. Let $G := (G_1, \dots, G_m)$ be centered Gaussian with covariance matrix C . If F is continuously differentiable with bounded first derivatives, then Gaussian integration by parts (see [19, Vol. 1, Sec. A.4]) gives

$$\mathbb{E}[G_i F(G)] = \sum_{j=1}^m C_{ij} \mathbb{E}[\partial_j F(G)].\tag{2.1}$$

We also use the following standard consequence of Gaussian concentration. Let G be a standard Gaussian vector, and let $\{F_v\}_{v \in I}$ be a finite set of L -Lipschitz functions on the vector space. Then, it holds that

$$\mathbb{E} \max_{v \in I} \{F_v(G) - \mathbb{E} F_v(G)\} \leq L \sqrt{2 \log |I|}.\tag{2.2}$$

Indeed, the Gaussian concentration inequality (see, for example, [16]) gives, for every $t > 0$ and every $v \in I$,

$$\mathbb{E} \exp(t(F_v(G) - \mathbb{E} F_v(G))) \leq \exp\left(\frac{t^2 L^2}{2}\right).$$

This inequality and Jensen's inequality imply

$$\begin{aligned}\mathbb{E} \max_{v \in I} (F_v(G) - \mathbb{E} F_v(G)) &\leq \frac{1}{t} \log \mathbb{E} \exp\left(t \max_{v \in I} (F_v(G) - \mathbb{E} F_v(G))\right) \\ &\leq \frac{1}{t} \log \sum_{v \in I} \mathbb{E} \exp(t(F_v(G) - \mathbb{E} F_v(G))) \\ &\leq \frac{\log |I|}{t} + \frac{t L^2}{2},\end{aligned}$$

where the first inequality follows from Jensen's inequality. For $|I| \geq 2$ and $L > 0$, optimize the upper bound by choosing $t = \sqrt{2 \log |I|}/L$. The cases $|I| = 1$ and $L = 0$ are immediate. Thus, we obtain (2.2).

2.2 The replica-symmetric path

Fix $(\beta, h, s) \in \mathcal{K} \times [0, 1]$. For $t \geq 0$, let $\mathbf{H}_t$ be the heat semigroup, i.e.

$$(\mathbf{H}_t \varphi)(x) := \mathbb{E} \varphi(x + \sqrt{t} Z).$$

Then, it holds that

$$\partial_t (\mathbf{H}_t \varphi)(x) = \frac{1}{2} \mathbf{H}_t (\partial^2 \varphi)(x). \quad (2.3)$$

For $0 \leq u \leq 1$ and $x \in \mathbb{R}$, define

$$\Psi(u, x) := \mathbb{E} \log \cosh \left(x + \beta \sqrt{s(q-u)_+} Z \right) + \frac{s\beta^2}{2} (1 - \max\{u, q\}), \quad Z \sim N(0, 1), \quad (2.4)$$

where $(a)_+ := \max\{a, 0\}$. Equivalently,

$$\Psi(u, x) = \begin{cases} (\mathbf{H}_{s\beta^2(q-u)} \log \cosh)(x) + \frac{s\beta^2}{2} (1 - q), & 0 \leq u \leq q, \\ \log \cosh(x) + \frac{s\beta^2}{2} (1 - u), & q \leq u \leq 1. \end{cases}$$

Differentiating (2.4) in x gives

$$\partial_x \Psi(u, x) = \mathbb{E} \tanh \left(x + \beta \sqrt{s(q-u)_+} Z \right) = \begin{cases} (\mathbf{H}_{s\beta^2(q-u)} \tanh)(x), & 0 \leq u \leq q, \\ \tanh(x), & q \leq u \leq 1. \end{cases} \quad (2.5)$$

For the upper interval, introduce the tilted heat semigroup

$$(\mathbf{T}_t \varphi)(x) := e^{-t/2} \frac{(\mathbf{H}_t(\varphi \cdot \cosh))(x)}{\cosh(x)}. \quad (2.6)$$

Differentiation under the Gaussian integral shows that its generator is $\frac{1}{2} \partial_{xx} + \tanh(x) \partial_x$. For each $u \in [0, 1]$, let $X_{s,u}$ denote a random variable whose law is specified by

$$\mathbb{E} \varphi(X_{s,u}) := \begin{cases} (\mathbf{H}_{\beta^2(1-s)q + s\beta^2 u} \varphi)(h), & 0 \leq u \leq q, \\ (\mathbf{H}_{\beta^2 q} (\mathbf{T}_{s\beta^2(u-q)} \varphi))(h), & q \leq u \leq 1. \end{cases} \quad (2.7)$$

In particular,

$$X_{s,q} \stackrel{d}{=} h + \beta \sqrt{q} Z. \quad (2.8)$$

The replica-symmetric free energy for $H_{N,s}$ along the path is

$$P_s^* := \log 2 + \mathbb{E} \Psi(0, h + \beta \sqrt{(1-s)q} Z) - \frac{s\beta^2}{2} \int_q^1 u \, du \quad (2.9)$$

$$= \log 2 + \mathbb{E} \log \cosh(h + \beta \sqrt{q} Z) + \frac{s\beta^2}{4} (1 - q)^2, \quad (2.10)$$

where we have used the semigroup property and $\beta^2(1-s)q + s\beta^2 q = \beta^2 q$.

Lemma 2.1. *If $s\beta^2(1-q) > 1$, then*

$$\beta^2 \mathbb{E} \operatorname{sech}^4(X_{s,u}) \leq \alpha, \quad q \leq u \leq 1. \quad (2.11)$$

Proof. Fix $q \leq u \leq 1$ and set

$$\sigma^2 := \beta^2 q, \quad t := s\beta^2(u - q), \quad X := h + \sigma Z.$$

By (2.7),

$$\mathbb{E} \operatorname{sech}^4(X_{s,u}) = \mathbb{E}[(\mathbb{T}_t \operatorname{sech}^4)(X)].$$

Thus it suffices to prove

$$\mathbb{E}[(\mathbb{T}_t \operatorname{sech}^4)(X)] \leq \mathbb{E} \operatorname{sech}^4(X).$$

We first note that

$$h < \sigma^2. \quad (2.12)$$

Indeed, suppose that $h \geq \sigma^2$. Since the convolution of an even function decreasing on $[0, \infty)$ with a centered Gaussian density has the same property,

$$1 - q = \mathbb{E} \operatorname{sech}^2(h + \sigma Z) \leq \mathbb{E} \operatorname{sech}^2(\sigma^2 + \sigma Z).$$

Let ε be a symmetric Rademacher variable, independent of Z , and put $Y = \sigma^2 \varepsilon + \sigma Z$. Then $\mathbb{E}[\varepsilon | Y] = \tanh Y$ due to $\mathbb{P}(\varepsilon = \pm 1, Y \in dy) = \frac{1}{2\sqrt{2\pi\sigma^2}} e^{-(y - (\pm\sigma^2))^2/2\sigma^2} dy$, and hence

$$\mathbb{E} \operatorname{sech}^2(\sigma^2 + \sigma Z) = \mathbb{E} \operatorname{Var}(\varepsilon | Y) \leq \inf_{a \in \mathbb{R}} \mathbb{E}(\varepsilon - aY)^2 = \frac{1}{1 + \sigma^2}.$$

Therefore

$$(1 - q)(1 + \beta^2 q) \leq 1,$$

which, since $q > 0$, implies $\beta^2(1 - q) \leq 1$. This contradicts $s\beta^2(1 - q) > 1$ because $s \leq 1$.

Now put

$$f := \operatorname{sech}^4, \quad g_t := \mathbb{T}_t f, \quad M(t) := \mathbb{E} g_t(X).$$

Since

$$g_t(x) = e^{-t/2} \frac{\mathbf{H}_t(\operatorname{sech}^3)(x)}{\cosh x},$$

g_t is even and nonincreasing on $[0, \infty)$; in particular,

$$g'_t(x) \leq 0, \quad x > 0.$$

By (2.3) and (2.6),

$$\partial_t(\mathbb{T}_t \varphi)(x) = \frac{1}{2 \cosh^2 x} \partial_x (\cosh^2 x \partial_x (\mathbb{T}_t \varphi)(x)). \quad (2.13)$$

Let $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ be the standard normal density. Since $X \sim N(h, \sigma^2)$, its density is $\sigma^{-1} \phi((x - h)/\sigma)$. From (2.13) and integration by parts in divergence form we have

$$\begin{aligned} M'(t) &= \frac{1}{2} \int_{\mathbb{R}} \partial_x (\cosh^2 x g'_t(x)) \frac{\phi((x - h)/\sigma)}{\sigma \cosh^2 x} dx \\ &= -\frac{1}{2} \int_{\mathbb{R}} \cosh^2 x g'_t(x) \partial_x \left(\frac{\phi((x - h)/\sigma)}{\sigma \cosh^2 x} \right) dx \\ &= \mathbb{E} \left[\left(\tanh X + \frac{X - h}{2\sigma^2} \right) g'_t(X) \right]. \end{aligned}$$

Pairing x and $-x$ gives

$$M'(t) = \int_0^\infty g'_t(x) K(x) dx, \quad (2.14)$$

where

$$K(x) := \frac{1}{\sigma} \phi\left(\frac{x-h}{\sigma}\right) \left(\tanh x + \frac{x-h}{2\sigma^2}\right) + \frac{1}{\sigma} \phi\left(\frac{-x-h}{\sigma}\right) \left(\tanh x + \frac{x+h}{2\sigma^2}\right).$$

Since

$$\frac{\phi((-x-h)/\sigma)}{\phi((x-h)/\sigma)} = e^{-2hx/\sigma^2},$$

we obtain

$$\frac{\sigma K(x)}{\phi((x-h)/\sigma) + \phi((-x-h)/\sigma)} = \tanh x + \frac{x}{2\sigma^2} - \frac{h}{2\sigma^2} \tanh\left(\frac{hx}{\sigma^2}\right).$$

Since $0 < h/\sigma^2 < 1$ (see (2.12)), $\tanh\left(\frac{hx}{\sigma^2}\right) \leq \tanh x$. This inequality and (1.4) implies that for $x > 0$

$$\frac{\sigma K(x)}{\phi((x-h)/\sigma) + \phi((-x-h)/\sigma)} \geq \left(1 - \frac{h}{2\sigma^2}\right) \tanh x + \frac{x}{2\sigma^2} > 0.$$

Thus $K(x) > 0$, while $g'_t(x) \leq 0$, and therefore (2.14) yields

$$M'(t) \leq 0.$$

In particular, it holds that $M(t) \leq M(0)$. From this inequality we obtain

$$\beta^2 \mathbb{E} \operatorname{sech}^4(X_{s,u}) \leq \beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta\sqrt{q} Z) = \alpha.$$

□

Proposition 2.2. *The function g_s defined by*

$$g_s(u) := \mathbb{E}[(\partial_x \Psi(u, X_{s,u}))^2] \quad (2.15)$$

satisfies

$$g_s(u) - u \begin{cases} > 0, & 0 \leq u < q, \\ = 0, & u = q, \\ < 0, & q < u \leq 1. \end{cases} \quad (2.16)$$

Moreover, there are constants $c_K, \varepsilon_K > 0$ such that

$$|g_s(u) - u| \geq c_K |u - q| \quad \text{whenever } |u - q| \leq \varepsilon_K. \quad (2.17)$$

Proof. Since (2.5) and (2.8) imply

$$g_s(q) = \mathbb{E}[\tanh^2(X_{s,q})] = q,$$

(2.17) holds for $u = q$.

We consider the case that $u < q$. Since (2.5) gives

$$g_s(u) = \mathbb{E}[(H_{s\beta^2(q-u)}(\tanh))(X_{s,u})^2],$$

by noting that (2.3) and (2.7) imply

$$\partial_u \mathbb{E} \varphi(X_{s,u}) = \frac{s\beta^2}{2} (H_{\beta^2(1-s)q+s\beta^2u}(\partial^2 \varphi))(h) = \frac{s\beta^2}{2} \mathbb{E}[(\partial^2 \varphi)(X_{s,u})],$$

we have

$$\begin{aligned}
g'_s(u) &= -s\beta^2 \partial_v \mathbb{E} [(\mathbf{H}_v(\tanh))(X_{s,u})^2] \big|_{v=s\beta^2(q-u)} + \partial_v \mathbb{E} [(\mathbf{H}_{s\beta^2(q-u)}(\tanh))(X_{s,v})^2] \big|_{v=u} \\
&= -s\beta^2 \mathbb{E} [(\mathbf{H}_{s\beta^2(q-u)}(\tanh))(X_{s,u})(\mathbf{H}_{s\beta^2(q-u)}(\partial^2 \tanh))(X_{s,u})] \\
&\quad + \frac{s\beta^2}{2} \mathbb{E} \left[\left(\partial_x^2 (\mathbf{H}_{s\beta^2(q-u)}(\tanh)(x))^2 \right) \big|_{x=X_{s,u}} \right] \\
&= s\beta^2 \mathbb{E} [(\mathbf{H}_{s\beta^2(q-u)}(\partial \tanh))(X_{s,u})^2].
\end{aligned}$$

Thus, we obtain

$$g'_s(u) = s\beta^2 \mathbb{E} \left[\left\{ (\mathbf{H}_{s\beta^2(q-u)}(\text{sech}^2))(X_{s,u}) \right\}^2 \right].$$

Jensen's inequality, the semigroup property and (2.7) imply

$$g'_s(u) \leq s\beta^2 \mathbb{E} [(\mathbf{H}_{s\beta^2(q-u)}(\text{sech}^4))(X_{s,u})] = s\beta^2 \mathbb{E} \text{sech}^4(h + \beta\sqrt{q}Z) = s\alpha.$$

Since $g_s(q) = q$, the integration of both sides implies

$$q - g_s(u) \leq s\alpha(q - u).$$

Hence, it holds that

$$g_s(u) - u \geq (1 - s\alpha)(q - u) \quad (0 \leq u < q). \quad (2.18)$$

This yields (2.16) for $0 \leq u < q$.

We consider the case that $u \geq q$. In this case, (2.5) gives $g_s(u) = \mathbb{E} \tanh^2(X_{s,u})$. On the other hand, (2.3), (2.6) and (2.7) imply

$$\begin{aligned}
\partial_u \mathbb{E} \varphi(X_{s,u}) &= \partial_u \mathbf{H}_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{\mathbf{H}_{s\beta^2(u-q)}(\varphi \cdot \cosh)}{\cosh} \right) (h) \\
&= -\frac{s\beta^2}{2} \mathbb{E} \varphi(X_{s,u}) + \frac{s\beta^2}{2} \mathbf{H}_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{\mathbf{H}_{s\beta^2(u-q)}(\partial^2(\varphi \cdot \cosh))}{\cosh} \right) (h) \\
&= \frac{s\beta^2}{2} \mathbb{E}(\partial^2 \varphi)(X_{s,u}) + s\beta^2 \mathbf{H}_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{\mathbf{H}_{s\beta^2(u-q)}((\partial \varphi) \cdot \sinh)}{\cosh} \right) (h).
\end{aligned}$$

Hence,

$$\begin{aligned}
g'_s(u) &= s\beta^2 \mathbb{E} \text{sech}^4(X_{s,u}) - 2s\beta^2 \mathbb{E} \tanh^2(X_{s,u}) \text{sech}^2(X_{s,u}) \\
&\quad + 2s\beta^2 \mathbf{H}_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{\mathbf{H}_{s\beta^2(u-q)}(\text{sech} \cdot \tanh^2)}{\cosh} \right) (h)
\end{aligned}$$

Thus, by recalling (2.6) and (2.7), we have

$$g'_s(u) = s\beta^2 \mathbb{E} \text{sech}^4(X_{s,u}), \quad g'_s(q) = s\alpha < 1. \quad (2.19)$$

If $s\beta^2(1 - q) > 1$, Lemma 2.1 gives $g'_s(u) \leq s\alpha < 1$, so $g_s(u) < u$ for $u > q$. We consider the case that $s\beta^2(1 - q) \leq 1$. Since $\text{sech}^4(x) \leq \text{sech}^2(x) = 1 - \tanh^2(x)$ for every $x \in \mathbb{R}$, by (2.19) we have

$$g'_s(u) \leq s\beta^2(1 - g_s(u)). \quad (2.20)$$

Then, (2.20) implies that for $u > q$

$$\partial_u(g_s(u) - u) \leq -s\beta^2(g_s(u) - u) + s\beta^2(1 - u) - 1 \leq -s\beta^2(g_s(u) - u).$$

Hence, by noting that (2.19) implies $g_s(u) - u < 0$ for $u > q$ sufficiently close to q and by applying Grönwall's inequality we have

$$g_s(u) - u < 0 \quad (u > q).$$

This yields (2.16) for $u > q$.

We have (2.17) from (2.18) and (2.19). $\square$

2.3 Two-replica Guerra–Talagrand coercivity

For an attainable overlap $v \in \mathcal{R}_N := \{-1, -1 + 2/N, \dots, 1\}$, define

$$Z_{N,s}^{(2)}(v) := \sum_{\substack{\sigma^1, \sigma^2 \in \Sigma_N \\ R_{12}=v}} e^{H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2)} \quad (2.21)$$

and retain the notation P_s^* from (2.10). For $\lambda \in \mathbb{R}$, define

$$f_\lambda(x_1, x_2) := \log \left(\frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 = \pm 1} \exp(\varepsilon_1 x_1 + \varepsilon_2 x_2 + \lambda \varepsilon_1 \varepsilon_2) \right). \quad (2.22)$$

For $(\beta, h, s) \in \mathcal{K} \times [0, 1]$, define ξ_s by

$$\xi_s(r) := \beta^2(1-s)qr + \frac{s\beta^2}{2}r^2. \quad (2.23)$$

Given $v \in [-1, 1]$, write $r_v := |v|$ and put $\iota_v := 1$ for $v \geq 0$ and $\iota_v := -1$ for $v < 0$. Let $\{a_j\}_{j=1}^{M+1}$ be the increasing list of the distinct elements of $\{0, q, r_v, 1\}$, i.e.

$$\{a_j\}_{j=1}^{M+1} = \{0, q, r_v, 1\} \quad \text{and} \quad a_j < a_{j+1} \quad (0 \leq j \leq M).$$

Note that $M = 1$ if $r_v = q, 1$, and $M = 2$ otherwise. Define

$$Q_j^v := \begin{pmatrix} a_j & \iota_v(a_j \wedge r_v) \\ \iota_v(a_j \wedge r_v) & a_j \end{pmatrix}, \quad 0 \leq j \leq M+1. \quad (2.24)$$

For $1 \leq j \leq M$, set

$$m_j := \begin{cases} 0, & a_j \leq q, \\ \frac{1}{2}, & q < a_j \leq r_v, \\ 1, & a_j > \max\{q, r_v\}. \end{cases} \quad (2.25)$$

The levels $\{m_j\}_{j=1}^M$ are nondecreasing. More explicitly, if $r_v \leq q$, $\{m_j\}_{j=1}^M = \{0, 1\}$; if $q < r_v < 1$, $\{m_j\}_{j=1}^M = \{0, \frac{1}{2}, 1\}$; and if $r_v = 1$, $\{m_j\}_{j=1}^M = \{0, \frac{1}{2}\}$. Thus $\{m_j\}_{j=1}^M \subset [0, 1]$ in every case, including the cases that $r_v = 0$ and that $r_v = q$. In the interpolation below we separately set $m_{M+1} = 1$, even in the case that $r_v = 1$.

Define, entrywise,

$$B_j := \beta^2(1-s)q \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + s\beta^2 Q_j^v = \xi_s'(Q_j^v), \quad 0 \leq j \leq M+1. \quad (2.26)$$

Since r_v is a breakpoint, every increment is of one of the forms

$$Q_j^v - Q_{j-1}^v = \begin{cases} (a_j - a_{j-1}) \begin{pmatrix} 1 & \iota_v \\ \iota_v & 1 \end{pmatrix}, & a_j \leq r_v, \\ (a_j - a_{j-1}) I_2, & a_{j-1} \geq r_v. \end{cases} \quad (2.27)$$

Both displayed matrices are positive semidefinite. Consequently,

$$B_j - B_{j-1} = s\beta^2(Q_j^v - Q_{j-1}^v) \succeq 0. \quad (2.28)$$

For a positive semidefinite 2×2 matrix C , let G_C be a centered Gaussian vector with covariance C , and define

$$\mathcal{T}_{m,C}F(x) := \begin{cases} m^{-1} \log \mathbb{E} e^{mF(x+G_C)}, & m > 0, \\ \mathbb{E} F(x + G_C), & m = 0. \end{cases} \quad (2.29)$$

Define deterministic stage functions by

$$U_{s,M}^{\lambda,v} = f_\lambda, \quad U_{s,j-1}^{\lambda,v} = \mathcal{T}_{m_j, B_j - B_{j-1}} U_{s,j}^{\lambda,v}, \quad (1 \leq j \leq M), \quad (2.30)$$

and for $x = (x_1, x_2) \in \mathbb{R}^2$ set

$$U_s^{\lambda,v}(x) := U_{s,0}^{\lambda,v}(x). \quad (2.31)$$

When a breakpoint should be clarified, we use the discrete notation

$$U_s^{\lambda,v}(a_j, x) := U_{s,j}^{\lambda,v}(x).$$

Equivalently, choose independent centered Gaussian vectors $G_1, \dots, G_M$ with $\text{Var}(G_j) = B_j - B_{j-1}$ and let $\mathcal{F}_j = \sigma(G_1, \dots, G_j)$, with $\mathcal{F}_0$ trivial. For fixed $x \in \mathbb{R}^2$, put

$$X_M^\lambda = f_\lambda \left(x + \sum_{k=1}^M G_k \right),$$

and for $0 \leq j \leq M$ define X_j recursively by

$$X_{j-1}^\lambda := \begin{cases} \frac{1}{m_j} \log \mathbb{E} \left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1} \right], & m_j > 0, \\ \mathbb{E} \left[X_j^\lambda \mid \mathcal{F}_{j-1} \right], & m_j = 0. \end{cases} \quad (2.32)$$

Then, it holds that

$$X_j^\lambda = U_{s,j}^{\lambda,v} \left(x + \sum_{k=1}^j G_k \right), \quad X_0^\lambda = U_s^{\lambda,v}(x). \quad (2.33)$$

In particular, $U_s^{\lambda,v}(x)$ is deterministic. This definition also covers the case that $s = 0$. In the case, every covariance increment vanishes.

Next, put

$$d_j := \frac{s\beta^2}{2} \sum_{k,\ell=1}^2 (Q_{j,k\ell}^v)^2, \quad 0 \leq j \leq M \quad (2.34)$$

where $Q_{j,k\ell}^v$ is the (k, ℓ) entry of Q_j^v . As a function of a breakpoint value a , this quantity is given by

$$d(a) = \begin{cases} 2s\beta^2 a^2, & 0 \leq a \leq r_v, \\ s\beta^2(a^2 + r_v^2), & r_v \leq a \leq 1. \end{cases}$$

Indeed, it holds that $d_j = d(a_j)$. Since $d(a)$ is continuous and nondecreasing,

$$d_j - d_{j-1} \geq 0. \quad (2.35)$$

Define

$$\mathcal{L}_s(v) := \frac{1}{2} \sum_{j=1}^M m_j(d_j - d_{j-1}). \quad (2.36)$$

This normalization is the scalar Gaussian contribution at the $t = 1$ endpoint of the interpolation in Appendix B, and is independent of v . Indeed, if $r_v \leq q$, then

$$\sum_{j=1}^M m_j(d_j - d_{j-1}) = d(1) - d(q) = s\beta^2(1 - q^2).$$

If $q < r_v < 1$,

$$\begin{aligned} \sum_{j=1}^M m_j(d_j - d_{j-1}) &= \frac{1}{2}(d(r_v) - d(q)) + d(1) - d(r_v) \\ &= s\beta^2(r_v^2 - q^2) + s\beta^2(1 - r_v^2) = s\beta^2(1 - q^2). \end{aligned}$$

If $r_v = 1$,

$$\sum_{j=1}^M m_j(d_j - d_{j-1}) = \frac{1}{2}(d(1) - d(q)) = s\beta^2(1 - q^2).$$

Therefore,

$$\mathcal{L}_s(v) = \frac{s\beta^2}{2}(1 - q^2), \quad (2.37)$$

in particular $\mathcal{L}_s(v)$ is independent of v .

With $Y_0 = h + \beta\sqrt{(1-s)q} Z_0$, set

$$\mathfrak{P}_s(\lambda, v) := 2 \log 2 + \mathbb{E} U_s^{\lambda, v}(Y_0, Y_0) - \lambda v - \mathcal{L}_s(v). \quad (2.38)$$

The following finite-volume Guerra–Talagrand upper bound is Lemma B.2, proved in Appendix B. It is a special case of the general two-dimensional bound of Talagrand and Chen [19, 13]:

$$\frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) \leq \inf_{\lambda \in \mathbb{R}} \mathfrak{P}_s(\lambda, v). \quad (2.39)$$

The following technical lemma is also postponed to Appendix C.

Lemma 2.3. *On every compact set on which λ is bounded, the maps*

$$(\beta, h, s, \lambda, v) \mapsto \mathfrak{P}_s(\lambda, v) - 2P_s^*, \quad (\beta, h, s, \lambda, v) \mapsto \partial_\lambda \mathfrak{P}_s(\lambda, v)$$

are jointly continuous for $(\beta, h, s, v) \in \mathcal{K} \times [0, 1] \times [-1, 1]$.

Lemma 2.4. *In the recursion (2.32), define the tilted conditional expectation by*

$$\mathbb{E}_j^\lambda[H] := \frac{\mathbb{E}\left[He^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1}\right]}{\mathbb{E}\left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1}\right]}, \quad m_j > 0,$$

and let $\mathbb{E}_j^\lambda[H] = \mathbb{E}[H \mid \mathcal{F}_{j-1}]$ when $m_j = 0$. Then

$$\partial_\lambda X_{j-1}^\lambda = \mathbb{E}_j^\lambda\left[\partial_\lambda X_j^\lambda\right], \quad (2.40)$$

$$\partial_{\lambda\lambda} X_{j-1}^\lambda = \mathbb{E}_j^\lambda\left[\partial_{\lambda\lambda} X_j^\lambda\right] + m_j \text{Var}_j^\lambda\left(\partial_\lambda X_j^\lambda\right). \quad (2.41)$$

Proof. For $m_j > 0$, recalling that

$$X_{j-1}^\lambda = \frac{1}{m_j} \log \mathbb{E}\left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1}\right],$$

we have

$$\begin{aligned} \partial_\lambda X_{j-1}^\lambda &= \frac{\mathbb{E}\left[\partial_\lambda X_j^\lambda e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1}\right]}{\mathbb{E}\left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1}\right]} = \mathbb{E}_j^\lambda[\partial_\lambda X_j^\lambda], \\ \partial_{\lambda\lambda} X_{j-1}^\lambda &= \mathbb{E}_j^\lambda[\partial_{\lambda\lambda} X_j^\lambda] + m_j \text{Var}_j^\lambda(\partial_\lambda X_j^\lambda). \end{aligned}$$

For $m_j = 0$, direct differentiation of the ordinary conditional expectation gives the same identities with no variance term. They complete the proof. $\square$

Corollary 2.5. *It holds*

$$\left|\partial_\lambda U_s^{\lambda,v}(x)\right| \leq 1, \quad 0 \leq \partial_{\lambda\lambda} U_s^{\lambda,v}(x) \leq \frac{5}{2}. \quad (2.42)$$

Consequently, uniformly in $(\beta, h, s, v, \lambda)$,

$$0 \leq \partial_{\lambda\lambda} \mathfrak{P}_s(\lambda, v) \leq \frac{5}{2}. \quad (2.43)$$

Proof. Let $\langle \cdot \rangle_{\lambda,x}$ denote expectation on $\{\pm 1\}^2$ with weights proportional to

$$\exp(\varepsilon_1 x_1 + \varepsilon_2 x_2 + \lambda \varepsilon_1 \varepsilon_2) \quad ((\varepsilon_1, \varepsilon_2) \in \{\pm 1\}^2).$$

Since $(\varepsilon_1 \varepsilon_2)^2 = 1$,

$$\partial_\lambda f_\lambda(x) = \langle \varepsilon_1 \varepsilon_2 \rangle_{\lambda,x}, \quad \partial_{\lambda\lambda} f_\lambda(x) = 1 - \langle \varepsilon_1 \varepsilon_2 \rangle_{\lambda,x}^2.$$

Hence, we have $|\partial_\lambda f_\lambda| \leq 1$ and $0 \leq \partial_{\lambda\lambda} f_\lambda \leq 1$. By applying (2.40) recursively, we have

$$\text{Var}_j^\lambda(\partial_\lambda X_j^\lambda) \leq \mathbb{E}_j^\lambda[(\partial_\lambda X_j^\lambda)^2] \leq 1.$$

Similarly, applying (2.41) recursively, we have

$$\partial_{\lambda\lambda} X_0^\lambda = \mathbb{E}_0^\lambda \cdots \mathbb{E}_M^\lambda \left[\partial_{\lambda\lambda} X_M^\lambda \right] + \sum_{j=0}^M m_j \mathbb{E}_0^\lambda \cdots \mathbb{E}_j^\lambda \left[\text{Var}_j^\lambda \left(\partial_\lambda X_j^\lambda \right) \right].$$

This equality, (2.33) and the fact that $0 \leq \partial_{\lambda\lambda} f_\lambda \leq 1$ yield

$$0 \leq \partial_{\lambda\lambda} U_s^{\lambda,v}(x) \leq 1 + \sum_{j=1}^M m_j. \quad (2.44)$$

If $r_v \leq q$, the sum is at most 1. If $q < r_v < 1$, $\{m_j\}_{j=1}^M = \{0, \frac{1}{2}, 1\}$ and their sum is $3/2$. If $r_v = 1$, $\{m_j\}_{j=1}^M = \{0, \frac{1}{2}\}$ and their sum is $1/2$. Noting that $M = 1, 2$, consequently we have in every boundary configuration,

$$1 + \sum_{j=1}^M m_j \leq \frac{5}{2}.$$

Together with (2.44), this proves (2.42).

Moreover, differentiation under the expectation in (2.38) gives

$$\partial_{\lambda\lambda} \mathfrak{P}_s(\lambda, v) = \mathbb{E} \partial_{\lambda\lambda} U_s^{\lambda,v}(Y_0, Y_0).$$

Hence (2.42) proves (2.43). □

Proposition 2.6. *Put*

$$\Gamma_s(v) := \inf_{\lambda \in \mathbb{R}} \mathfrak{P}_s(\lambda, v).$$

There is a constant $c_K > 0$ such that

$$\Gamma_s(v) \leq 2P_s^* - c_K(v - q)^2 \quad (2.45)$$

for $(\beta, h, s, v) \in \mathcal{K} \times [0, 1] \times [-1, 1]$. Consequently, for every N and $v \in \mathcal{R}_N$,

$$\frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) \leq 2P_s^* - c_K(v - q)^2. \quad (2.46)$$

Moreover,

$$\Gamma_s(q) = 2P_s^*. \quad (2.47)$$

Proof. We first evaluate the finite recursion at $\lambda = 0$. Suppose that $-q \leq v \leq q$. The level-1 recursion above q (What is this?) and the identities

$$f_0(x_1, x_2) = \log \cosh x_1 + \log \cosh x_2, \quad \partial_\lambda f_\lambda(x_1, x_2)|_{\lambda=0} = \tanh x_1 \tanh x_2$$

give, at the breakpoint q ,

$$\begin{aligned} U_s^{0,v}(q, x_1, x_2) &= s\beta^2(1 - q) + \log \cosh x_1 + \log \cosh x_2, \\ \partial_\lambda U_s^{\lambda,v}(q, x_1, x_2)|_{\lambda=0} &= \tanh x_1 \tanh x_2. \end{aligned}$$

All recursions below q have level zero. (What does it mean?) Let $(Y_1(v), Y_2(v))$ a Gaussian pair such that both $Y_1(v)$ and $Y_2(v)$ have means h and variances $\beta^2 q$, and have the covariance

$$\text{Cov}(Y_1(v), Y_2(v)) = \beta^2((1 - s)q + sv).$$

It follows directly from the conditional expectations below q (What does this mean?), (2.37) and (2.10) that

$$\mathfrak{P}_s(0, v) = 2P_s^*, \quad \partial_\lambda \mathfrak{P}_s(0, v) = \tilde{g}_s(v) - v \quad (-q \leq v \leq q) \quad (2.48)$$

where

$$\tilde{g}_s(v) := \mathbb{E}[\tanh Y_1(v) \tanh Y_2(v)].$$

Gaussian differentiation (**What is this?**) and the Cauchy–Schwarz inequality yield

$$\tilde{g}'_s(v) = s\beta^2 \mathbb{E}[\operatorname{sech}^2 Y_1(v) \operatorname{sech}^2 Y_2(v)] \leq s\alpha$$

where α is defined in (1.3). Since $\operatorname{Cov}(Y_1(q), Y_2(q)) = \beta^2 q = \operatorname{Var}(Y_0)$, it holds that $\tilde{g}_s(q) = \mathbb{E}[\tanh^2(Y_0)] = q$ where $Y_0 = h + \beta\sqrt{(1-s)q}Z_0$. Hence, it follows that

$$\tilde{g}_s(v) - v \geq (1 - s\alpha)(q - v) > 0, \quad -q \leq v < q. \quad (2.49)$$

For $0 \leq v \leq q$, put $t = s\beta^2(q - v)$, let $X \sim N(h, \beta^2((1-s)q + sv))$, and let Z_1 and Z_2 be random variables with the standard normal law which are also independent of other random variables. The variables

$$X + \sqrt{t}Z_1, \quad X + \sqrt{t}Z_2,$$

have the joint law of $(Y_1(v), Y_2(v))$, and are independent under the probability measure conditioned on X . Consequently,

$$\tilde{g}_s(v) = \mathbb{E} \left[(\mathbf{H}_t(\tanh)(X))^2 \right] = g_s(v), \quad 0 \leq v \leq q \quad (2.50)$$

by (2.5), (2.7), and (2.15).

Now let $q < v \leq 1$ and put $t = s\beta^2(v - q)$. Above v the level is 1 and the coordinate increments are independent. The stage recursion gives

$$\begin{aligned} U_s^{0,v}(v, x_1, x_2) &= s\beta^2(1 - v) + \log \cosh x_1 + \log \cosh x_2, \\ D_v(x_1, x_2) &:= \partial_\lambda U_s^{\lambda,v}(v, x_1, x_2) \Big|_{\lambda=0} = \tanh x_1 \tanh x_2. \end{aligned}$$

For $v = 1$, the stage above v is absent and these formulas reduce to the terminal identities. Between q and v , the increment is $(\sqrt{t}Z, \sqrt{t}Z)$ and its level is $1/2$. Hence, on the diagonal,

$$\begin{aligned} U_s^{0,v}(q, (x, x)) &= s\beta^2(1 - v) + 2 \log \mathbb{E} \cosh(x + \sqrt{t}Z) \\ &= 2 \log \cosh x + s\beta^2(1 - q), \end{aligned} \quad (2.51)$$

where $\mathbb{E} \cosh(x + \sqrt{t}Z) = e^{t/2} \cosh x$. The derivative recursion in Lemma 2.4 gives

$$\begin{aligned} D_q(x, x) &= \partial_\lambda U_s^{\lambda,v}(q, (x, x)) \Big|_{\lambda=0} \\ &= \frac{\mathbb{E}[\tanh^2(x + \sqrt{t}Z) \cosh(x + \sqrt{t}Z)]}{\mathbb{E} \cosh(x + \sqrt{t}Z)} = (\mathbf{T}_t \tanh^2)(x). \end{aligned} \quad (2.52)$$

Below q , the increment is common to both coordinates and its level is zero. Combining it with Y_0 in (2.51) gives

$$\mathbb{E} U_s^{0,v}(Y_0, Y_0) = 2\mathbb{E} \log \cosh(h + \beta\sqrt{q}Z) + s\beta^2(1 - q).$$

Likewise, (2.52) and (2.7) give

$$\mathbb{E} \partial_\lambda U_s^{\lambda,v}(Y_0, Y_0) \Big|_{\lambda=0} = (\mathbf{H}_{\beta^2 q} \mathbf{T}_{s\beta^2(v-q)} \tanh^2)(h) = g_s(v).$$

Thus, together with (2.48) and (2.50), we obtain

$$\mathfrak{P}_s(0, v) = 2P_s^*, \quad \partial_\lambda \mathfrak{P}_s(0, v) = g_s(v) - v, \quad 0 \leq v \leq 1. \quad (2.53)$$

We also need a strict gap in the case that $v < -q$. Suppose first that $s > 0$ and write

$$r = |v| > q, \quad t = s\beta^2(r - q) > 0.$$

The stage recursion above r , whose level is 1, gives

$$U_s^{0,v}(r, x_1, x_2) = s\beta^2(1 - r) + \log \cosh x_1 + \log \cosh x_2.$$

The stage recursion from r to q , whose level is $1/2$ and whose Gaussian increment is $(\sqrt{t}Z, -\sqrt{t}Z)$, gives

$$U_s^{0,v}(q, x_1, x_2) = s\beta^2(1 - r) + 2 \log \mathbb{E} \sqrt{\cosh(x_1 + \sqrt{t}Z) \cosh(x_2 - \sqrt{t}Z)}.$$

By Cauchy–Schwarz,

$$\begin{aligned} U_s^{0,v}(q, x_1, x_2) &\leq s\beta^2(1 - r) + t + \log \cosh x_1 + \log \cosh x_2 \\ &= \Psi(q, x_1) + \Psi(q, x_2). \end{aligned}$$

The formulas remain valid at $v = -1$, where the stage above r has zero covariance. Since $t > 0$, equality holds only when $x_1 = -x_2$. Below q the recursion is linear, and it evaluates the two sides at

$$X_1 = Y_0 + \beta\sqrt{s\bar{q}}Z, \quad X_2 = Y_0 - \beta\sqrt{s\bar{q}}Z.$$

Since $\mathbb{P}(X_1 = -X_2) = \mathbb{P}(Y_0 = 0) = 0$, also when $s = 1$ because then $Y_0 = h > 0$, the propagated inequality is strict. Therefore

$$\mathfrak{P}_s(0, v) < 2P_s^*, \quad v < -q, \quad s > 0. \quad (2.54)$$

At $s = 0$, a direct calculation gives

$$\mathfrak{P}_0(0, v) = 2P_0^*, \quad \partial_\lambda \mathfrak{P}_0(0, v) = q - v, \quad v < -q. \quad (2.55)$$

We now prove the desired bound by applying the results above. Choose $\varepsilon_K < q_K/2$ so that (2.17) holds when $|v - q| \leq \varepsilon_K$. Proposition 2.2 gives $a_K > 0$ such that

$$(g_s(v) - v)(v - q) \leq -a_K(v - q)^2, \quad |v - q| \leq \varepsilon_K. \text{ (Why this one is needed?)} \quad (2.56)$$

These values of v are positive, so (2.43) and (2.53) imply

$$\mathfrak{P}_s(\lambda, v) \leq 2P_s^* + \lambda(g_s(v) - v) + \frac{M_K}{2}\lambda^2.$$

Taking $\lambda = a_K(v - q)/M_K$ yields

$$\Gamma_s(v) \leq 2P_s^* - \frac{a_K^2}{2M_K}(v - q)^2, \quad |v - q| \leq \varepsilon_K. \quad (2.57)$$

It remains to consider overlaps bounded away from q . We claim that

$$\Gamma_s(v) < 2P_s^* \quad \text{for } v \neq q. \quad (2.58)$$

In the case that $0 \leq v < q$ and $q < v \leq 1$, this follows from (2.53) and (2.16) by moving λ slightly in the direction opposite to the derivative. For the case $-q \leq v < 0$, use (2.48) and (2.49). For the

case that $v < -q$ and $s > 0$, apply (2.54). For the case that $v < -q$ and $s = 0$, (2.55) gives the claim by taking a sufficiently small negative λ .

Consider the compact set

$$\mathcal{D}_\mathcal{K} = \{(\beta, h, s, v) : (\beta, h) \in \mathcal{K}, 0 \leq s \leq 1, -1 \leq v \leq 1, |v - q| \geq \varepsilon_\mathcal{K}\}.$$

At every point $\theta \in \mathcal{D}_\mathcal{K}$, (2.58) provides $\lambda_\theta \in \mathbb{R}$ such that

$$\mathfrak{P}_s(\lambda_\theta, v) < 2P_s^*.$$

In view of Lemma 2.3, this inequality persists with a positive margin on a neighborhood of θ . The compactness of $\mathcal{D}_\mathcal{K}$ gives $\delta_\mathcal{K} > 0$ such that

$$\Gamma_s(v) \leq 2P_s^* - \delta_\mathcal{K}, \quad |v - q| \geq \varepsilon_\mathcal{K}.$$

Since $|v - q| \leq 2$,

$$\Gamma_s(v) \leq 2P_s^* - \frac{\delta_\mathcal{K}}{4}(v - q)^2, \quad |v - q| \geq \varepsilon_\mathcal{K}. \quad (2.59)$$

Combining (2.57) and (2.59) proves (2.45), with

$$c_\mathcal{K} = \min \left\{ \frac{a_\mathcal{K}^2}{2M_\mathcal{K}}, \frac{\delta_\mathcal{K}}{4} \right\} > 0.$$

The finite-volume estimate (2.46) follows from (2.39).

Finally, at $v = q$, (2.53) gives

$$\mathfrak{P}_s(0, q) = 2P_s^*, \quad \partial_\lambda \mathfrak{P}_s(0, q) = g_s(q) - q = 0.$$

Since (2.43) implies that $\lambda \mapsto \mathfrak{P}_s(\lambda, q)$ is convex, $\lambda = 0$ is a global minimizer. Hence

$$\Gamma_s(q) = 2P_s^*,$$

which proves (2.47). □

3 Preliminary overlap concentration

3.1 Finite-volume Gronwall closure

For $\rho \geq 0$, define the coupled pressure $p_{N,s}^{(2)}(\rho)$ and its normalized excess $F_{N,s}(\rho)$ by

$$p_{N,s}^{(2)}(\rho) := \frac{1}{2N} \mathbb{E} \log \sum_{\sigma^1, \sigma^2} \exp \left(H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2) + \frac{\rho N}{2} Q_{12}^2 \right), \quad (3.1)$$

$$F_{N,s}(\rho) := p_{N,s}^{(2)}(\rho) - \phi_N(s) = \frac{1}{2N} \mathbb{E} \log \left\langle e^{\rho N Q_{12}^2/2} \right\rangle_s, \quad (3.2)$$

respectively. Fix

$$0 < \rho_0 < 2c_\mathcal{K}, \quad (3.3)$$

where $c_\mathcal{K}$ is the coercivity constant in Proposition 2.6.

Lemma 3.1. *It holds*

$$p_{N,s}^{(2)}(\rho_0) \leq P_s^* + O_\mathcal{K}(1) \sqrt{\frac{\log(N+1)}{N}}. \quad (3.4)$$

Proof. Let $L_v := \log Z_{N,s}^{(2)}(v)$. Each L_v is a Lipschitz function of the Gaussian disorder with squared Lipschitz constant at most $O_{\mathcal{K}}(1)N$. Indeed,

$$|\partial_{g_{ij}} L_v| \leq \frac{\sqrt{2}\beta\sqrt{s}}{\sqrt{N}}, \quad |\partial_{z_i} L_v| \leq 2\beta\sqrt{(1-s)q},$$

and summing up the squares over i, j and i gives the claim. Hence, by (2.2),

$$\mathbb{E} \max_{v \in \mathcal{R}_N} (L_v - \mathbb{E} L_v) \leq O_{\mathcal{K}}(1) \sqrt{N \log(N+1)} \quad (3.5)$$

where we recall $\mathcal{R}_N := \{-1, -1 + 2/N, \dots, 1\}$. Thus, by using (3.5), (2.46) and (3.3), we have

$$\begin{aligned} 2Np_{N,s}^{(2)}(\rho_0) &= \mathbb{E} \log \sum_{v \in \mathcal{R}_N} \exp \left(\mathbb{E} L_v + (L_v - \mathbb{E} L_v) + \frac{\rho_0 N}{2} (v - q)^2 \right) \\ &\leq \log(N+1) + \max_v \left\{ \mathbb{E} L_v + \frac{\rho_0 N}{2} (v - q)^2 \right\} + O_{\mathcal{K}}(1) \sqrt{N \log(N+1)} \\ &\leq 2NP_s^* + O_{\mathcal{K}}(1) \sqrt{N \log(N+1)}. \end{aligned}$$

This proves (3.4). $\square$

Put

$$D_N(s) := P_s^* - \phi_N(s). \quad (3.6)$$

Lemma 3.2. *It holds that for $s \in [0, 1]$*

$$D_N(s) \leq O_{\mathcal{K}}(1) \sqrt{\frac{\log(N+1)}{N}}, \quad A_s \leq O_{\mathcal{K}}(1) \sqrt{\frac{\log(N+1)}{N}}. \quad (3.7)$$

Proof. Write

$$\epsilon_N := O_{\mathcal{K}}(1) \sqrt{\frac{\log(N+1)}{N}}.$$

Similarly to [15, (14)], via Gaussian integration by parts (see [19, Vol. 1, Sec. A.4]), it follows that

$$\phi'_N(s) = \frac{\beta^2}{4} \left((1-q)^2 - \nu_s[Q_{12}^2] \right). \quad (3.8)$$

Combining this identity with (2.10) and the definition of A_s in (1.9) yields

$$D'_N(s) = \frac{\beta^2}{4} A_s. \quad (3.9)$$

On the other hand, convexity of $F_{N,s}(\rho)$ in $\rho \in [0, \infty)$, where $F_{N,s}(\rho)$ is defined in (3.2), gives

$$\frac{1}{4} A_s = \partial_{\rho} F_{N,s}(\rho) \big|_{\rho=0+} \leq \frac{F_{N,s}(\rho_0)}{\rho_0}. \quad (3.10)$$

By (3.4),

$$F_{N,s}(\rho_0) \leq D_N(s) + \epsilon_N. \quad (3.11)$$

Consequently,

$$D'_N(s) \leq \frac{\beta^2}{\rho_0} (D_N(s) + \epsilon_N). \quad (3.12)$$

Gronwall's inequality, the compactness of $\mathcal{K}$, the continuity of $D_N(s)$ in $s \in [0, 1]$, and the fact that $D_N(0) = 0$ imply the inequality for $D_N(s)$ in (3.7). The inequality for A_s in (3.7) follows from this inequality, (3.10) and (3.11). $\square$

3.2 Fixed-deviation overlap concentration

Corollary 3.3. *For every $\varepsilon > 0$, there are constants $c_{\mathcal{K},\varepsilon}, C_{\mathcal{K},\varepsilon} > 0$ such that*

$$\sup_{\substack{(\beta,h) \in \mathcal{K} \\ 0 \leq s \leq 1}} \nu_s[|Q_{12}| \geq \varepsilon] \leq C_{\mathcal{K},\varepsilon} e^{-c_{\mathcal{K},\varepsilon} N}. \quad (3.13)$$

Proof. Let

$$Y_{N,s} := \log \left\langle e^{\rho_0 N Q_{12}^2 / 2} \right\rangle_s.$$

By (3.2), (3.7) and (3.11),

$$\mathbb{E} Y_{N,s} = 2N F_{N,s}(\rho_0) \leq O_{\mathcal{K}}(1) N \epsilon_N = o(N) \quad (3.14)$$

uniformly in $s \in [0, 1]$. It is known that the map

$$\begin{aligned} ((g_{ij})_{i,j}, (Z_i)_i) &\mapsto Y_{N,s} = \log \sum_{\sigma^1, \sigma^2} \exp \left(H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2) + \frac{\rho_0 N}{2} Q_{12}^2 \right) \\ &\quad - \log \sum_{\sigma^1, \sigma^2} \exp (H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2)), \end{aligned}$$

where we note that $H_{N,s}$ depends on $((g_{ij})_{i,j}, (Z_i)_i)$, is Lipschitz continuous with Lipschitz constant $C_{\mathcal{K}} \sqrt{N}$ (see [19, Theorem 1.3.5] and the discussion after the theorem). The Gaussian concentration inequality (see [19, Theorem 1.3.4]) therefore gives, for all large enough N ,

$$\mathbb{P}(|Y_{N,s} - \mathbb{E} Y_{N,s}| > t) \leq \exp(-N^{-1} t^2 / C_{\mathcal{K}}) \quad (t \in [0, \infty)).$$

This inequality and (3.14) yield

$$\mathbb{P}\left(Y_{N,s} > \frac{\rho_0 \varepsilon^2 N}{4}\right) \leq C_{\mathcal{K},\varepsilon} e^{-c_{\mathcal{K},\varepsilon} N}. \quad (3.15)$$

For each realization of the disorder, on $\{|Q_{12}| \geq \varepsilon\}$ we have $\mathbf{1}_{\{|Q_{12}| \geq \varepsilon\}} \leq \exp\left(\frac{\rho_0 N}{2}(Q_{12}^2 - \varepsilon^2)\right)$. Therefore,

$$\langle \mathbf{1}_{\{|Q_{12}| \geq \varepsilon\}} \rangle_s \leq \exp\left(-\frac{\rho_0 \varepsilon^2 N}{2}\right) \left\langle \exp\left(\frac{\rho_0 N Q_{12}^2}{2}\right) \right\rangle_s = \exp\left(-\frac{\rho_0 \varepsilon^2 N}{2} + Y_{N,s}\right).$$

Hence, (3.15) implies

$$\begin{aligned} \nu_s[|Q_{12}| \geq \varepsilon] &\leq \mathbb{E} \left[\exp\left(-\frac{\rho_0 \varepsilon^2 N}{2} + Y_{N,s}\right) \mathbf{1}_{\{Y_{N,s} \leq \frac{\rho_0 \varepsilon^2 N}{4}\}} \right] + \mathbb{P}\left(Y_{N,s} > \frac{\rho_0 \varepsilon^2 N}{4}\right) \\ &\leq \exp\left(-\frac{\rho_0 \varepsilon^2 N}{4}\right) + \mathbb{P}\left(Y_{N,s} > \frac{\rho_0 \varepsilon^2 N}{4}\right). \end{aligned}$$

Therefore, we obtain (3.13). $\square$

3.3 Cavity closure

Recall from (1.9) that

$$A_s = \nu_s[Q_{12}^2], \quad B_s = \nu_s[Q_{12}Q_{13}], \quad C_s = \nu_s[Q_{12}Q_{34}].$$

Introduce the three cavity modes

$$U_s := A_s - 4B_s + 3C_s, \quad V_s := 2B_s - 3C_s, \quad D_s := A_s - 2B_s + C_s, \quad (3.16)$$

and put

$$\kappa := 1 - 4q + 3r, \quad \zeta := 2q + q^2 - 3r. \quad (3.17)$$

Proposition 3.4. *There are remainders $\mathcal{R}_{N,s}^U, \mathcal{R}_{N,s}^V, \mathcal{R}_{N,s}^D$ such that*

$$(1 - s\beta^2\kappa)U_s = \frac{\kappa}{N} + \mathcal{R}_{N,s}^U, \quad (3.18)$$

$$(1 - s\beta^2\kappa)V_s - s\beta^2\zeta U_s = \frac{\zeta}{N} + \mathcal{R}_{N,s}^V, \quad (3.19)$$

$$(1 - s\alpha)D_s = \frac{\alpha}{\beta^2 N} + \mathcal{R}_{N,s}^D, \quad (3.20)$$

and uniformly on $\mathcal{K} \times [0, 1]$,

$$\max_{X \in \{U, V, D\}} |\mathcal{R}_{N,s}^X| \leq C_{\mathcal{K}}(N^{-3/2} + \nu_s[|Q_{12}|^3]). \quad (3.21)$$

The proof follows from a straightforward, though technical, application of Talagrand's cavity argument; see [19, Chapter 1]. We postpone the details to Appendix A.

Since $r \leq q$,

$$\alpha - \beta^2\kappa = 2\beta^2(q - r) \geq 0.$$

Thus $\beta^2\kappa \leq \alpha < 1$. If $\kappa < 0$, then $1 - s\beta^2\kappa \geq 1$; if $\kappa \geq 0$, then $1 - s\beta^2\kappa \geq 1 - \alpha$. Hence (1.5) gives

$$1 - s\alpha \geq \delta_{\mathcal{K}}, \quad 1 - s\beta^2\kappa \geq \delta_{\mathcal{K}} \quad (3.22)$$

for all $(\beta, h, s) \in \mathcal{K} \times [0, 1]$. Equations (3.18)–(3.20), in this order, and the boundedness of ζ on $\mathcal{K}$ imply

$$|U_s| + |V_s| + |D_s| \leq C_{\mathcal{K}} \left(\frac{1}{N} + N^{-3/2} + \nu_s[|Q_{12}|^3] \right).$$

Since

$$A_s = -V_s - 2U_s + 3D_s,$$

we conclude that

$$A_s \leq \frac{C_{\mathcal{K}}}{N} + C_{\mathcal{K}}\nu_s[|Q_{12}|^3]. \quad (3.23)$$

4 Conclusions

4.1 Absorption of the cubic remainder

Fix $\varepsilon > 0$. Since $|Q_{12}| \leq 2$,

$$\begin{aligned} \nu_s[|Q_{12}|^3] &\leq \varepsilon \nu_s[Q_{12}^2] + 8 \nu_s[|Q_{12}| > \varepsilon] \\ &\leq \varepsilon A_s + 8C_{\mathcal{K},\varepsilon} e^{-c_{\mathcal{K},\varepsilon} N}, \end{aligned} \quad (4.1)$$

where Corollary 3.3 was used. Choose $\varepsilon = \varepsilon_{\mathcal{K}}$ so that the coefficient of A_s obtained by substituting (4.1) into (3.23) is at most $1/2$. We obtain

$$A_s = \nu_s[Q_{12}^2] \leq \frac{C_{\mathcal{K}}}{N}$$

uniformly on $\mathcal{K} \times [0, 1]$, after increasing $C_{\mathcal{K}}$ to include the finitely many small values of N . This proves the first claim of Theorem 1.1.

4.2 Free energy

Recall that $D_N(s) := P_s^* - \phi_N(s)$. The identity (3.9), $D'_N(s) = \frac{\beta^2}{4}\nu_s[Q_{12}^2]$, with $P_1^* = \phi^{\text{RS}}(\beta, h)$ and $D_N(0) = 0$ gives

$$\phi^{\text{RS}}(\beta, h) - \phi_N(1) = D_N(1) = \frac{\beta^2}{4} \int_0^1 \nu_s[Q_{12}^2] \, ds. \quad (4.2)$$

The integrand is nonnegative, and the bound above makes the right-hand side at most $C_{\mathcal{K}}/N$. This proves the free-energy claim of Theorem 1.1.

4.3 Replicon susceptibility

The $O(N^{-1})$ second-moment estimate and (4.1) imply, for every fixed $\varepsilon > 0$,

$$\sup_{\mathcal{K} \times [0, 1]} N \nu_s[|Q_{12}|^3] \leq C_{\mathcal{K}}\varepsilon + 8NC_{\mathcal{K}, \varepsilon} e^{-c_{\mathcal{K}, \varepsilon} N}.$$

First let $N \rightarrow \infty$ and then $\varepsilon \downarrow 0$. Thus

$$\nu_s[|Q_{12}|^3] = o_{\mathcal{K}}(N^{-1})$$

uniformly in s . Equation (3.20), together with (3.21) and (3.22), now gives

$$ND_s = \frac{\alpha}{\beta^2(1 - s\alpha)} + o_{\mathcal{K}}(1),$$

which is the final claim of Theorem 1.1.

A Proof of the cavity approximations

Proof of Proposition 3.4. All implicit constants in this proof depend only on $\mathcal{K}$. Write

$$\eta_{N,s} := N^{-3/2} + \nu_s[|Q_{12}|^3], \quad \varepsilon_a := \sigma_N^a, \quad Q_{ab}^- := \frac{1}{N} \sum_{i < N} \sigma_i^a \sigma_i^b - q,$$

so that

$$Q_{ab} = Q_{ab}^- + \frac{1}{N} \varepsilon_a \varepsilon_b. \quad (\text{A.1})$$

By site exchangeability after averaging over the disorder,

$$A_s = \nu_s[Q_{12}(\varepsilon_1 \varepsilon_2 - q)], \quad B_s = \nu_s[Q_{12}(\varepsilon_1 \varepsilon_3 - q)], \quad C_s = \nu_s[Q_{12}(\varepsilon_3 \varepsilon_4 - q)].$$

Use the last-spin cavity interpolation. Write $\sigma = (\rho, \varepsilon)$, with $\rho \in \{-1, 1\}^{N-1}$, and replace the interaction of the last spin with the first $N - 1$ spins by

$$\frac{\beta\sqrt{su}}{\sqrt{N}} \sum_{i < N} \frac{g_{iN} + g_{Ni}}{\sqrt{2}} \rho_i \varepsilon + \beta\sqrt{s(1-u)q} \widehat{z} \varepsilon, \quad 0 \leq u \leq 1,$$

where $\widehat{z}$ is an independent standard Gaussian. Denote the corresponding averaged Gibbs expectation by $\nu_{s,u}$. Then $\nu_{s,1} = \nu_s$. At $u = 0$, the last spin is independent of the first $N - 1$ spins and is subject to the field

$$h + \beta\sqrt{(1-s)q} z_N + \beta\sqrt{sq} \widehat{z} \stackrel{d}{=} h + \beta\sqrt{q} Z.$$

Thus, for distinct replica indices,

$$\mathbb{E}_0[\varepsilon_a \varepsilon_b] = q, \quad \mathbb{E}_0[\varepsilon_a \varepsilon_b \varepsilon_c \varepsilon_d] = r. \quad (\text{A.2})$$

Gaussian integration by parts gives, for every bounded function F of n replicas,

$$\begin{aligned} \frac{d}{du} \nu_{s,u}[F] &= s\beta^2 \left\{ \sum_{1 \leq a < b \leq n} \nu_{s,u}[F \varepsilon_a \varepsilon_b Q_{ab}^-] - n \sum_{a=1}^n \nu_{s,u}[F \varepsilon_a \varepsilon_{n+1} Q_{a,n+1}^-] \right. \\ &\quad \left. + \frac{n(n+1)}{2} \nu_{s,u}[F \varepsilon_{n+1} \varepsilon_{n+2} Q_{n+1,n+2}^-] \right\}. \end{aligned} \quad (\text{A.3})$$

Indeed, the derivative of the covariance of the interpolating last-spin field is $s\beta^2 \varepsilon \varepsilon' Q^-(\rho, \rho')$, while its diagonal value is independent of the configuration.

Set $M_3(u) := \nu_{s,u}[|Q_{12}^-|^3]$. Formula (A.3) and $|Q_{ab}^-| \leq 2$ give

$$|M'_3(u)| \leq C_K M_3(u).$$

Since

$$|Q_{12}^-|^3 \leq 4|Q_{12}|^3 + \frac{4}{N^3},$$

Gronwall's inequality, applied backward from $u = 1$, yields

$$\sup_{0 \leq u \leq 1} \nu_{s,u}[|Q_{12}^-|^3] \leq C_K \eta_{N,s}. \quad (\text{A.4})$$

For

$$P^- = Q_{ab}^- Q_{cd}^-, \quad (ab, cd) \in \{(12, 12), (12, 13), (12, 34)\},$$

the derivative of $\nu_{s,u}[P^-]$ is a finite linear combination of expectations of three cavity overlaps. Hence Hölder's inequality and (A.4) show that

$$|\nu_{s,0}[P^-] - \nu_{s,1}[P^-]| \leq C_K \eta_{N,s}.$$

Moreover, (A.1) gives

$$|P^- - Q_{ab} Q_{cd}| \leq \frac{|Q_{ab}| + |Q_{cd}|}{N} + \frac{1}{N^2}.$$

Using

$$\frac{|x|}{N} \leq \frac{|x|^3}{3} + \frac{2}{3N^{3/2}},$$

we obtain

$$|\nu_{s,0}[P^-] - \nu_s[Q_{ab}Q_{cd}]| \leq C_K \eta_{N,s}. \quad (\text{A.5})$$

Define

$$A_s^- := \nu_{s,0}[(Q_{12}^-)^2], \quad B_s^- := \nu_{s,0}[Q_{12}^- Q_{13}^-], \quad C_s^- := \nu_{s,0}[Q_{12}^- Q_{34}^-],$$

and form U_s^-, V_s^-, D_s^- from A_s^-, B_s^-, C_s^- as in (3.16). It follows from (A.5) that

$$|U_s^- - U_s| + |V_s^- - V_s| + |D_s^- - D_s| \leq C_K \eta_{N,s}. \quad (\text{A.6})$$

Next write

$$\begin{aligned} A_s &= \nu_s[G_A] + \frac{1}{N} \nu_s[E_A], \\ B_s &= \nu_s[G_B] + \frac{1}{N} \nu_s[E_B], \\ C_s &= \nu_s[G_C] + \frac{1}{N} \nu_s[E_C], \end{aligned} \quad (\text{A.7})$$

where

$$\begin{aligned} G_A &:= Q_{12}^-(\varepsilon_1 \varepsilon_2 - q), & E_A &:= \varepsilon_1 \varepsilon_2 (\varepsilon_1 \varepsilon_2 - q), \\ G_B &:= Q_{12}^-(\varepsilon_1 \varepsilon_3 - q), & E_B &:= \varepsilon_1 \varepsilon_2 (\varepsilon_1 \varepsilon_3 - q), \\ G_C &:= Q_{12}^-(\varepsilon_3 \varepsilon_4 - q), & E_C &:= \varepsilon_1 \varepsilon_2 (\varepsilon_3 \varepsilon_4 - q). \end{aligned}$$

Put

$$g_j(u) := \nu_{s,u}[G_j], \quad e_j(u) := \nu_{s,u}[E_j], \quad j \in \{A, B, C\},$$

and take the same linear combinations as in (3.16):

$$g_U := g_A - 4g_B + 3g_C, \quad g_V := 2g_B - 3g_C, \quad g_D := g_A - 2g_B + g_C,$$

with analogous definitions for e_U, e_V, e_D .

Independence at $u = 0$ gives $g_j(0) = 0$. Applying (A.3) twice, Hölder's inequality and (A.4) give

$$g_X(1) = g'_X(0) + C_K \eta_{N,s}, \quad X \in \{U, V, D\}. \quad (\text{A.8})$$

For a replica edge $\{a, b\}$, (A.2) implies

$$\mathbb{E}_0[(\varepsilon_p \varepsilon_q - q) \varepsilon_a \varepsilon_b] = \begin{cases} 1 - q^2, & \{a, b\} = \{p, q\}, \\ q - q^2, & \{a, b\} \text{ shares one endpoint with } \{p, q\}, \\ r - q^2, & \{a, b\} \cap \{p, q\} = \emptyset. \end{cases} \quad (\text{A.9})$$

Substitution into (A.3), followed directly by the three linear combinations above, gives

$$\begin{aligned} g'_U(0) &= s\beta^2 \kappa U_s^-, \\ g'_V(0) &= s\beta^2 (\zeta U_s^- + \kappa V_s^-), \\ g'_D(0) &= s\alpha D_s^-. \end{aligned} \quad (\text{A.10})$$

This is the cancellation for which the modes in (3.16) were chosen.

The last-spin moments also give

$$\begin{aligned} e_U(0) &= (1 - q^2) - 4(q - q^2) + 3(r - q^2) = \kappa, \\ e_V(0) &= 2(q - q^2) - 3(r - q^2) = \zeta, \\ e_D(0) &= 1 - 2q + r = \frac{\alpha}{\beta^2}. \end{aligned} \tag{A.11}$$

Since E_A, E_B, E_C are uniformly bounded, (A.3) yields

$$|e'_X(u)| \leq C_K \nu_{s,u}[|Q_{12}^-|], \quad X \in \{U, V, D\}.$$

The inequality

$$\frac{|Q_{12}^-|}{N} \leq \frac{|Q_{12}^-|^3}{3} + \frac{2}{3N^{3/2}}$$

and (A.4) therefore imply

$$\frac{1}{N} |e_X(1) - e_X(0)| \leq C_K \eta_{N,s}, \quad X \in \{U, V, D\}. \tag{A.12}$$

Taking the three mode combinations in (A.7), and then using (A.6), (A.8), (A.10), (A.11), and (A.12), gives

$$\begin{aligned} U_s &= s\beta^2 \kappa U_s + \frac{\kappa}{N} + C_K \eta_{N,s}, \\ V_s &= s\beta^2 (\zeta U_s + \kappa V_s) + \frac{\zeta}{N} + C_K \eta_{N,s}, \\ D_s &= s\alpha D_s + \frac{\alpha}{\beta^2 N} + C_K \eta_{N,s}. \end{aligned}$$

These are (3.18)–(3.21). □

B Finite-step proof of the specialized Guerra–Talagrand bound

B.1 A finite-step differentiation formula

The following lemma isolates the only hierarchical Gaussian calculation needed in the proof.

Lemma B.1. *Let $\mathcal{S}$ be finite, let μ be a nonzero finite measure on $\mathcal{S}$, and fix*

$$0 = m_0 \leq m_1 \leq \dots \leq m_M \leq m_{M+1} = 1.$$

Let $\{G_{k,t}(\xi) : \xi \in \mathcal{S}, t \in [0, 1]\}_{0 \leq k \leq M}$ be independent centered Gaussian processes with almost surely C^1 sample paths in $t \in (0, 1)$, and let $K_{k,t}$ denote their covariance kernels. Assume that $K_{k,t}$ is bounded. Put

$$\mathcal{G}_t(\xi) := \sum_{k=0}^M G_{k,t}(\xi), \quad C_t^{(j)}(\xi, \eta) := \sum_{k=0}^j K_{k,t}(\xi, \eta),$$

and write $C_t = C_t^{(M)}$. For a deterministic function $V : \mathcal{S} \rightarrow \mathbb{R}$, define

$$X_M(t) = \log \sum_{\xi \in \mathcal{S}} \mu(\xi) e^{\mathcal{G}_t(\xi) + V(\xi)}$$

and, recursively for $j = M, \dots, 1$,

$$X_{j-1}(t) := \begin{cases} \frac{1}{m_j} \log \mathbb{E}_j e^{m_j X_j(t)}, & m_j > 0, \\ \mathbb{E}_j X_j(t), & m_j = 0, \end{cases}$$

where $\mathbb{E}_j$ integrates only the field $G_{j,t}$. For $0 \leq j \leq M$, set

$$\mathcal{I}_j A := \frac{\mathbb{E}_j [A e^{m_j X_j(t)}]}{\mathbb{E}_j e^{m_j X_j(t)}},$$

and let

$$p_t(\xi) := \frac{\mu(\xi) e^{\mathcal{G}_t(\xi) + V(\xi)}}{\sum_{\eta \in \mathcal{S}} \mu(\eta) e^{\mathcal{G}_t(\eta) + V(\eta)}}.$$

For deterministic functions F and $0 \leq j \leq M$, define

$$\begin{aligned} \mathbb{E} \langle F(\xi) \rangle_t &:= \sum_{\xi \in \mathcal{S}} F(\xi) \mathcal{I}_0 \cdots \mathcal{I}_M p_t(\xi), \\ \mathbb{E} \langle F(\xi, \eta) \rangle_{j,t} &:= \sum_{\xi, \eta \in \mathcal{S}} F(\xi, \eta) \mathcal{I}_0 \cdots \mathcal{I}_j \left[(\mathcal{I}_{j+1} \cdots \mathcal{I}_M p_t(\xi)) (\mathcal{I}_{j+1} \cdots \mathcal{I}_M p_t(\eta)) \right], \end{aligned}$$

where an empty operator product is the identity. Then, for $0 < t < 1$,

$$\frac{d}{dt} \mathbb{E}_0 X_0(t) = \frac{1}{2} \mathbb{E} \left\langle \dot{C}_t(\xi, \xi) \right\rangle_t - \frac{1}{2} \sum_{j=0}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \dot{C}_t^{(j)}(\xi, \eta) \right\rangle_{j,t} \quad (\text{B.1})$$

where $\dot{C}_t^{(j)}(\xi, \eta) := \frac{\partial}{\partial t} C_t^{(j)}(\xi, \eta)$.

Proof. Throughout the proof, for fixed t , we regard $X_j(t)$ as a smooth function of the Gaussian coordinates

$$\{G_{k,t}(\xi) : 0 \leq k \leq j, \xi \in \mathcal{S}\}$$

that have not yet been integrated out; the fields at levels $j+1, \dots, M$ have already been averaged out by the recursion. Thus $\partial_x X_{j-1} = \mathcal{I}_j(\partial_x X_j)$ whenever x is a Gaussian coordinate not integrated by $\mathbb{E}_j$ or the time coordinate t . Indeed, if $Z_j = \mathbb{E}_j e^{m_j X_j}$ and $m_j > 0$, then

$$\partial_x X_{j-1} = \frac{1}{m_j} \frac{\partial_x Z_j}{Z_j} = \frac{\mathbb{E}_j [e^{m_j X_j} \partial_x X_j]}{Z_j} = \mathcal{I}_j(\partial_x X_j). \quad (\text{B.2})$$

The same identity for $m_j = 0$ follows by differentiating $X_{j-1} = \mathbb{E}_j X_j$. From (B.2) it follows that

$$\partial_{G_{k,t}(\xi)} X_k = \mathcal{I}_{k+1} \cdots \mathcal{I}_M p_t(\xi). \quad (\text{B.3})$$

For any differentiable A , the quotient rule gives the more general identity

$$\partial_y \mathcal{I}_j A = \mathcal{I}_j(\partial_y A) + m_j \mathcal{I}_j(A \partial_y X_j) - m_j \mathcal{I}_j(A) \mathcal{I}_j(\partial_y X_j). \quad (\text{B.4})$$

Consequently,

$$\partial_{xy} X_{j-1} = \mathcal{I}_j(\partial_{xy} X_j) + m_j \{ \mathcal{I}_j(\partial_x X_j \partial_y X_j) - \mathcal{I}_j(\partial_x X_j) \mathcal{I}_j(\partial_y X_j) \}. \quad (\text{B.5})$$

Again, this includes $m_j = 0$, since the last two terms then vanish.

By (B.2), we have

$$\begin{aligned}
\frac{d}{dt} \mathbb{E} X_0 &= \mathcal{I}_0 \dots \mathcal{I}_M \frac{d}{dt} X_M(t) \\
&= \sum_{k=0}^M \mathcal{I}_0 \dots \mathcal{I}_M \left(\sum_{\xi \in \mathcal{S}} p_t(\xi) \dot{G}_{k,t}(\xi) \right) \\
&= \sum_{k=0}^M \sum_{\xi \in \mathcal{S}} \mathcal{I}_0 \dots \mathcal{I}_k \left(\dot{G}_{k,t}(\xi) (\mathcal{I}_{k+1} \dots \mathcal{I}_M (p_t(\xi))) \right).
\end{aligned} \tag{B.6}$$

When $m_k > 0$, by Gaussian integration by parts (see [19, Vol. 1, Sec. A.4]), i.e. for centered jointly Gaussians X, Y_i ,

$$\mathbb{E}[X f(Y_1 \dots Y_d)] = \sum_{i=1}^d \text{Cov}(X, Y_i), \mathbb{E}[\partial_i f(Y)],$$

where $\text{Cov}(\dot{G}_{k,t}(\xi), \dot{G}_{k,t}(\eta)) = \dot{K}_{k,t}(\xi, \eta)$, by (B.3) we have

$$\begin{aligned}
\mathcal{I}_k \left(\dot{G}_{k,t}(\xi) (\mathcal{I}_{k+1} \dots \mathcal{I}_M p_t(\xi)) \right) &= Z_k^{-1} \mathbb{E}_k \left[e^{m_k X_k} \dot{G}_{k,t}(\xi) (\mathcal{I}_{k+1} \dots \mathcal{I}_M p_t(\xi)) \right] \\
&= \sum_{\eta \in \mathcal{S}} \dot{K}_{k,t}(\xi, \eta) (Z_k)^{-1} \mathbb{E}_k \left[\partial_{G_{k,t}(\eta)} (e^{m_k X_k} (\mathcal{I}_{k+1} \dots \mathcal{I}_M p_t(\xi))) \right] \\
&= \sum_{\eta \in \mathcal{S}} \dot{K}_{k,t}(\xi, \eta) (m_k Z_k)^{-1} \mathbb{E}_k \left[\partial_{G_{k,t}(\eta)} \partial_{G_{k,t}(\xi)} e^{m_k X_k} \right].
\end{aligned}$$

Let $H_{j,t}(\xi) = \sum_{r=0}^j G_{r,t}(\xi)$. For fixed k , write

$$u_k^\xi = \partial_{H_{k,t}(\xi)} X_k, \quad h_k^{\xi\eta} = \partial_{H_{k,t}(\xi)} \partial_{H_{k,t}(\eta)} X_k.$$

The contribution obtained by differentiating only the covariance of $G_{k,t}$ is

$$(m_k Z_k)^{-1} \mathbb{E}_k [\partial_{G_k(\xi)} \partial_{G_k(\eta)} e^{m_k X_k}] = \mathcal{I}_k \left(h_k^{\xi\eta} + m_k u_k^\xi u_k^\eta \right).$$

For $m_k = 0$, the same formula follows directly from (B.6), with the second summand equal to zero. At $k = 0$ it also holds because $\mathcal{I}_0 = \mathbb{E}_0$ and $m_0 = 0$. Hence, letting

$$\mathcal{P}_k = \mathcal{I}_0 \mathcal{I}_1 \dots \mathcal{I}_k, \quad Q_k^{\xi\eta} = h_k^{\xi\eta} + m_k u_k^\xi u_k^\eta,$$

by (B.6) we have

$$\frac{d}{dt} \mathbb{E} X_0 = \frac{1}{2} \sum_{k=0}^M \sum_{\xi, \eta \in \mathcal{S}} \dot{K}_{k,t}(\xi, \eta) \mathcal{P}_k Q_k^{\xi\eta}. \tag{B.7}$$

It remains to compute $\mathcal{P}_k Q_k^{\xi\eta}$. At the terminal level, direct differentiation gives

$$u_M^\xi = p_t(\xi), \quad h_M^{\xi\eta} = \mathbf{1}_{\{\xi=\eta\}} p_t(\xi) - p_t(\xi) p_t(\eta).$$

Since $m_{M+1} = 1$, it follows that

$$Q_M^{\xi\eta} = \mathbf{1}_{\{\xi=\eta\}} p_t(\xi) - (m_{M+1} - m_M) u_M^\xi u_M^\eta. \tag{B.8}$$

Note that (B.3) implies $\partial_{H_{j-1,t}(\xi)} X_j = \partial_{H_{j,t}(\xi)} X_j$. For $j = M, \dots, 1$, equations (B.2) and (B.5) yield

$$u_{j-1}^\xi = \mathcal{I}_j u_j^\xi, \quad h_{j-1}^{\xi\eta} = \mathcal{I}_j h_j^{\xi\eta} + m_j \left(\mathcal{I}_j (u_j^\xi u_j^\eta) - u_{j-1}^\xi u_{j-1}^\eta \right),$$

and hence

$$Q_{j-1}^{\xi\eta} = h_{j-1}^{\xi\eta} + m_{j-1} u_{j-1}^\xi u_{j-1}^\eta = \mathcal{I}_j Q_j^{\xi\eta} - (m_j - m_{j-1}) u_{j-1}^\xi u_{j-1}^\eta.$$

By applying $\mathcal{P}_{j-1}$ to both sides of this equality, we have

$$\mathcal{P}_{j-1} Q_{j-1}^{\xi\eta} = \mathcal{P}_j Q_j^{\xi\eta} - (m_j - m_{j-1}) \mathcal{P}_{j-1} (u_{j-1}^\xi u_{j-1}^\eta).$$

Iterating from $j = M$ down to $j = k + 1$ and using (B.8) gives

$$\mathcal{P}_k Q_k^{\xi\eta} = \mathcal{P}_M (\mathbf{1}_{\{\xi=\eta\}} p_t(\xi)) - \sum_{j=k}^M (m_{j+1} - m_j) \mathcal{P}_j (u_j^\xi u_j^\eta). \quad (\text{B.9})$$

Note that similarly to (B.3) it holds that $u_j^\xi = \mathcal{I}_{j+1} \cdots \mathcal{I}_M p_t(\xi)$. Substituting (B.9) into (B.7) therefore give

$$\begin{aligned} \frac{d}{dt} \mathbb{E} X_0 &= \frac{1}{2} \mathbb{E} \left\langle \sum_{k=0}^M \dot{K}_{k,t}(\xi, \xi) \right\rangle_t - \frac{1}{2} \sum_{k=0}^M \sum_{j=k}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \dot{K}_{k,t}(\xi, \eta) \right\rangle_{j,t} \\ &= \frac{1}{2} \mathbb{E} \left\langle \dot{C}_t(\xi, \xi) \right\rangle_t - \frac{1}{2} \sum_{j=0}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \dot{C}_t^{(j)}(\xi, \eta) \right\rangle_{j,t}, \end{aligned}$$

where we have used $\sum_{k=0}^M \sum_{j=k}^M \dot{K}_{k,t}(\xi, \eta) = \sum_{j=0}^M \sum_{k=0}^j \dot{K}_{k,t}(\xi, \eta) = \sum_{j=0}^M \dot{C}_t^{(j)}(\xi, \eta)$. $\square$

B.2 Application to the two-replica path

Fix $N \geq 1$, $\beta > 0$, $h \in \mathbb{R}$, $s \in [0, 1]$, and $q \in (0, 1)$. We use the same notation as in Section 2.3. Let $H_{N,s}^{\text{cen}}$ be a centered Gaussian field on Σ_N normalized by

$$\mathbb{E} H_{N,s}^{\text{cen}}(\sigma) H_{N,s}^{\text{cen}}(\tau) = N \xi_s(R(\sigma, \tau)). \quad (\text{B.10})$$

Set

$$H_{N,s}(\sigma) = H_{N,s}^{\text{cen}}(\sigma) + h \sum_{i=1}^N \sigma_i$$

and

$$\mathcal{R}_N = \{-1, -1 + 2/N, \dots, 1\}.$$

For $v \in \mathcal{R}_N$, let

$$Z_{N,s}^{(2)}(v) = \sum_{\substack{\sigma^1, \sigma^2 \in \Sigma_N \\ R(\sigma^1, \sigma^2) = v}} e^{H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2)}. \quad (\text{B.11})$$

We prove (2.39) as follows.

Lemma B.2. *For every $N \geq 1$ and $v \in \mathcal{R}_N$,*

$$\frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) \leq \inf_{\lambda \in \mathbb{R}} \mathfrak{P}_s(\lambda, v). \quad (\text{B.12})$$

Proof. In view of (2.28) and (2.35), we are able to choose independent Gaussian vectors $z_{i,j} \in \mathbb{R}^2$ and independent Gaussian variables y_j with

$$\mathbb{E} z_{i,j} z_{i,j}^\top = B_j - B_{j-1}, \quad \mathbb{E} y_j^2 = d_j - d_{j-1}.$$

Let $Y_{i,0}$ be independent copies of $Y_0 = h + \beta\sqrt{(1-s)q} Z$. Since

$$B_0 = \beta^2(1-s)q \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

the vector $(Y_{i,0} - h, Y_{i,0} - h)$ supplies the level-zero covariance B_0 .

Write

$$\mathcal{S}_N(v) = \{(\sigma^1, \sigma^2) \in \Sigma_N^2 : R(\sigma^1, \sigma^2) = v\}.$$

For $\sigma = (\sigma^1, \sigma^2) \in \mathcal{S}_N(v)$, introduce the independent Gaussian random variables $\{G_{j,t}(\sigma)\}_{j=0}^M$ with parameter $t \in [0, 1]$ as

$$\begin{aligned} G_{0,t}(\sigma) &:= \sqrt{t}(H_{N,s}^{\text{cen}}(\sigma^1) + H_{N,s}^{\text{cen}}(\sigma^2)) + \sqrt{1-t} \sum_{i=1}^N \sum_{a=1}^2 (Y_{i,0} - h) \sigma_i^a, \\ G_{j,t}(\sigma) &:= \sqrt{1-t} \sum_{i=1}^N \sum_{a=1}^2 z_{i,j}^a \sigma_i^a + \sqrt{tN} y_j, \quad 1 \leq j \leq M, \end{aligned}$$

and the deterministic term

$$V(\sigma) := h \sum_{i=1}^N (\sigma_i^1 + \sigma_i^2) + \lambda \sum_{i=1}^N \sigma_i^1 \sigma_i^2 - \lambda N v.$$

Let

$$X_M(t) := \log \sum_{\sigma \in \mathcal{S}_N(v)} \exp \left(\sum_{j=0}^M G_{j,t}(\sigma) + V(\sigma) \right),$$

define $\{X_j(t)\}_{j=0}^M$ as in Lemma B.1, and set

$$\varphi_N(t) := \frac{1}{N} \mathbb{E}_0 X_0(t).$$

We first compute the values at the endpoints. At $t = 0$, remove the overlap constraint from the partition function. The maps in (2.30) are monotone, and the independent site increments factorize. The defining recursion **What is this?** therefore gives

$$\varphi_N(0) \leq 2 \log 2 + \mathbb{E} U_s^{\lambda,v}(Y_0, Y_0) - \lambda v. \quad (\text{B.13})$$

The factor $2 \log 2$ comes from the normalization $1/4$ in f_λ .

At $t = 1$, the Lagrange term vanishes on $\mathcal{S}_N(v)$, while the sitewise Gaussian increments vanish. For a quantity x independent of y_j , the j th recursion gives

$$\begin{cases} \frac{1}{m_j} \log \mathbb{E}_j e^{m_j(x + \sqrt{N} y_j)} = x + \frac{N m_j}{2} (d_j - d_{j-1}), & m_j > 0, \\ \mathbb{E}_j(x + \sqrt{N} y_j) = x, & m_j = 0. \end{cases}$$

Hence, by applying this equality recursively we have ($m_M = 1$? Is the label should be shifted?)

$$\begin{aligned}\varphi_N(1) &= \frac{1}{N} E_0 \dots E_{M-1} \left[\log \mathbb{E}_M e^{m_M X_M(1)} \right] + \frac{1}{2} \sum_{j=1}^M m_j (d_j - d_{j-1}) \\ &= \frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) + \mathcal{L}_s(v)\end{aligned}\tag{B.14}$$

where $Z_{N,s}^{(2)}(v)$ and $\mathcal{L}_s(v)$ are defined in (B.11) and (2.36), respectively.

It remains to compute the derivative of φ_N . Set $m_0 = 0$, $m_{M+1} = 1$, and for two spin pairs $\sigma, \tau \in \mathcal{S}_N(v)$ write

$$q_j^{k\ell} = Q_{j,k\ell}^v, \quad R^{k\ell} = R(\sigma^k, \tau^\ell).$$

Also write

$$\theta_s(x) = x \xi'_s(x) - \xi_s(x) = \frac{s\beta^2}{2} x^2.$$

Denote the covariance of $\sum_{i=0}^j G_{i,t}(\sigma)$ and $\sum_{i=0}^j G_{i,t}(\tau)$ by $C_t^{(j)}(\sigma, \tau)$. If $\sigma_i = \tau_i$ for $i = 1, 2, \dots, j$, then it holds that

$$C_t^{(j)}(\sigma, \tau) = tN \sum_{k,\ell=1}^2 \xi_s(R^{k\ell}) + (1-t)N \sum_{k,\ell=1}^2 \xi'_s(q_j^{k\ell}) R^{k\ell} + tN \sum_{k,\ell=1}^2 \theta_s(q_j^{k\ell}) \text{Why?} \tag{B.15}$$

$$= N \sum_{k,\ell=1}^2 \xi'_s(q_j^{k\ell}) R^{k\ell} + t \left(\frac{s\beta^2 N}{2} \sum_{k,\ell=1}^2 (R^{k\ell} - q_j^{k\ell})^2 \right), \tag{B.16}$$

due to $\xi_s(x) - \xi_s(y) - \xi'_s(y)(x - y) = \frac{s\beta^2}{2}(x - y)^2$. For a single constrained pair, its 2×2 overlap matrix is

$$(R(\sigma^k, \sigma^\ell))_{k,\ell=1}^2 = \begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix} = Q_M^v.$$

Consequently,

$$\dot{C}_t^{(M)}(\sigma, \sigma) = \frac{s\beta^2 N}{2} \sum_{k,\ell=1}^2 (Q_{M,k\ell}^v - Q_{M,k\ell}^v)^2 = 0.$$

Hence, Lemma B.1 now yields

$$\varphi'_N(t) = -\frac{s\beta^2}{4} \sum_{j=0}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \sum_{k,\ell=1}^2 (R^{k\ell} - Q_{j,k\ell}^v)^2 \right\rangle_{j,t} \leq 0. \tag{B.17}$$

Combining (B.13), (B.14), and (B.17), and taking the infimum over λ proves (B.12). $\square$

C Proof of a technical lemma

Proof of Lemma 2.3. Fix $L < \infty$ and let $(\beta_n, h_n, s_n, \lambda_n, v_n) \rightarrow (\beta, h, s, \lambda, v)$, with $|\lambda_n| \leq L$. Write $q_n = q(\beta_n, h_n)$ and $r_n = |v_n|$. By the continuity of q established after (1.1), $q_n \rightarrow q$, while $r_n \rightarrow r := |v|$. There are only finitely many possible orders of q_n and r_n and signs of v_n , so it is enough to work on a subsequence on which these choices are fixed.

For each n , the definition of $U_{s_n}^{\lambda_n, v_n}$ is a composition of at most three operators from (2.29). On a stage whose limiting breakpoint length is positive, its level is eventually fixed and the covariance

$B_{j,n} - B_{j-1,n}$ converges to its limiting covariance. A stage whose breakpoints coalesce has covariance tending to zero, and its operator converges to the identity even if its level changes. This accounts for $v = 0$, $|v| = q$, and $|v| = 1$. If $s = 0$, all the covariances vanish, so the same conclusion is immediate.

On $|\lambda| \leq L$, the terminal data obey

$$|f_\lambda(x_1, x_2)| \leq |x_1| + |x_2| + L, \quad |\partial_\lambda f_\lambda| \leq 1.$$

For $m > 0$, differentiation of (2.29) gives

$$\partial_\lambda(\mathcal{T}_{m,C}F) = \frac{\mathbb{E}[(\partial_\lambda F)(x + G_C)e^{mF(x+G_C)}]}{\mathbb{E}e^{mF(x+G_C)}},$$

with the ordinary Gaussian expectation when $m = 0$. Hence every step preserves $|\partial_\lambda F| \leq 1$. The analogous coordinate derivative formula and $|\partial_{x_a} f_\lambda| \leq 1$ give a common local equicontinuity bound. For any convergent sequence of stage covariances $C_n \rightarrow C$, couple $G_{C_n} = C_n^{1/2}Z$. Dominated convergence shows that each operator and its λ -derivative converge locally uniformly. The same argument applies when $C = 0$, independently of the value of $m_{j,n}$. Iterating over the finite composition gives local uniform convergence of both $U_{s_n}^{\lambda_n, v_n}(\cdot)$ and its λ -derivative.

Finally, as $n \rightarrow \infty$

$$Y_{0,n} = h_n + \beta_n \sqrt{(1 - s_n)q_n} Z \longrightarrow Y_0 = h + \beta \sqrt{(1 - s)q} Z \quad \text{in } L^1,$$

and the sequence $(Y_{0,n})$ is uniformly integrable. The local uniform convergence above and the common linear-growth bound therefore give convergence after evaluation at $(Y_{0,n}, Y_{0,n})$. Since (2.37) holds and P_s^* is continuous, the definition (2.38) and

$$\partial_\lambda \mathfrak{P}_s(\lambda, v) = \mathbb{E} \partial_\lambda U_s^{\lambda, v}(Y_0, Y_0) - v$$

prove the asserted joint continuity. □

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Statement on the Use of Artificial Intelligence

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